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(54) **BIT LINE AND WORD LINE CONNECTION
FOR MEMORY ARRAY**

(58) **Field of Classification Search**

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G11C 13/0004; H10B 63/20;

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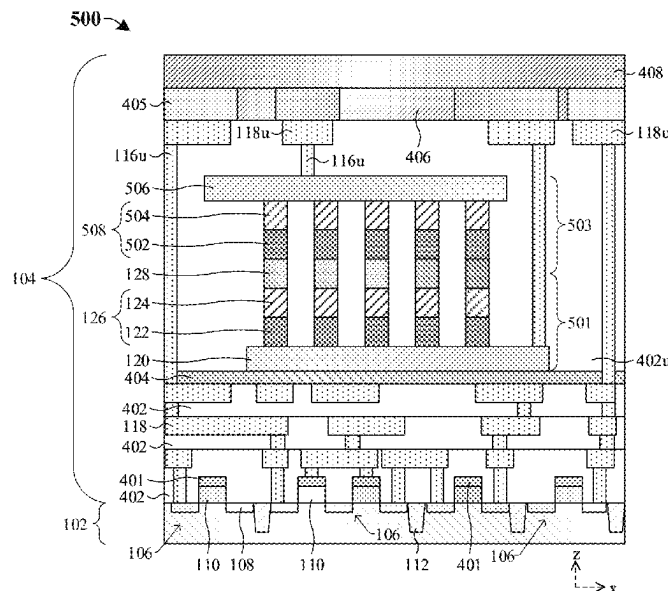
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ABSTRACT

Various embodiments of the present application are directed towards an integrated chip including a first conductive interconnect structure overlying a substrate. A first memory stack is disposed on the first conductive interconnect structure. A second conductive interconnect structure overlies the first memory stack. The second conductive interconnect structure is spaced laterally between opposing sidewalls of the first conductive interconnect structure. A third conductive interconnect structure is disposed on the first conductive interconnect structure. A top surface of the third conductive interconnect structure is vertically above the second conductive interconnect structure.

20 Claims, 11 Drawing Sheets



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(58) **Field of Classification Search**

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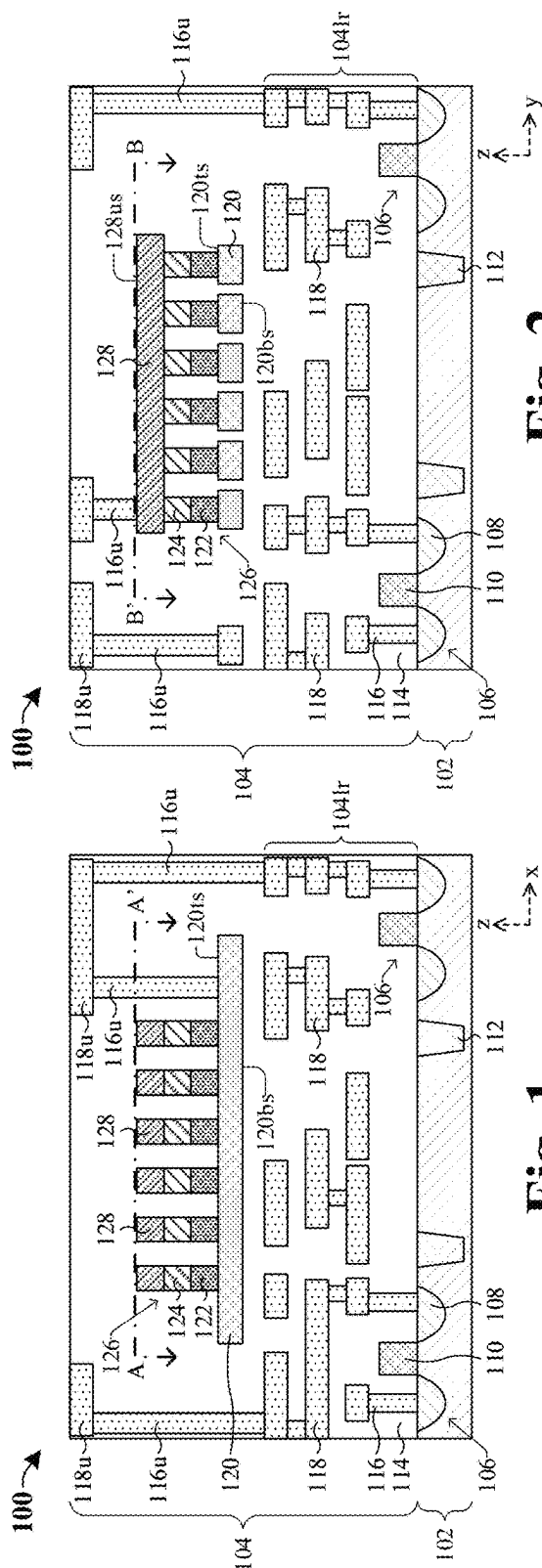


Fig. 1

Fig. 2

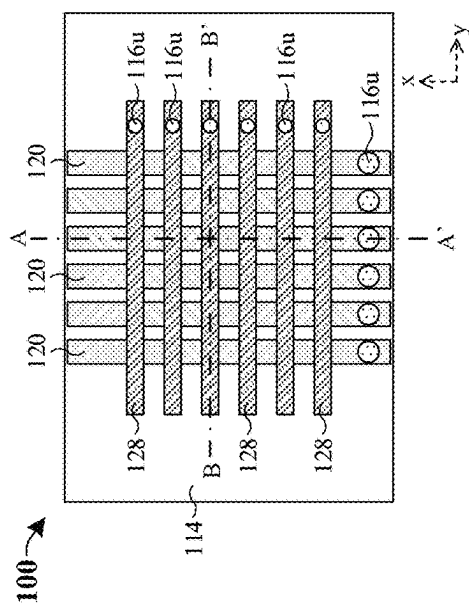


Fig. 3

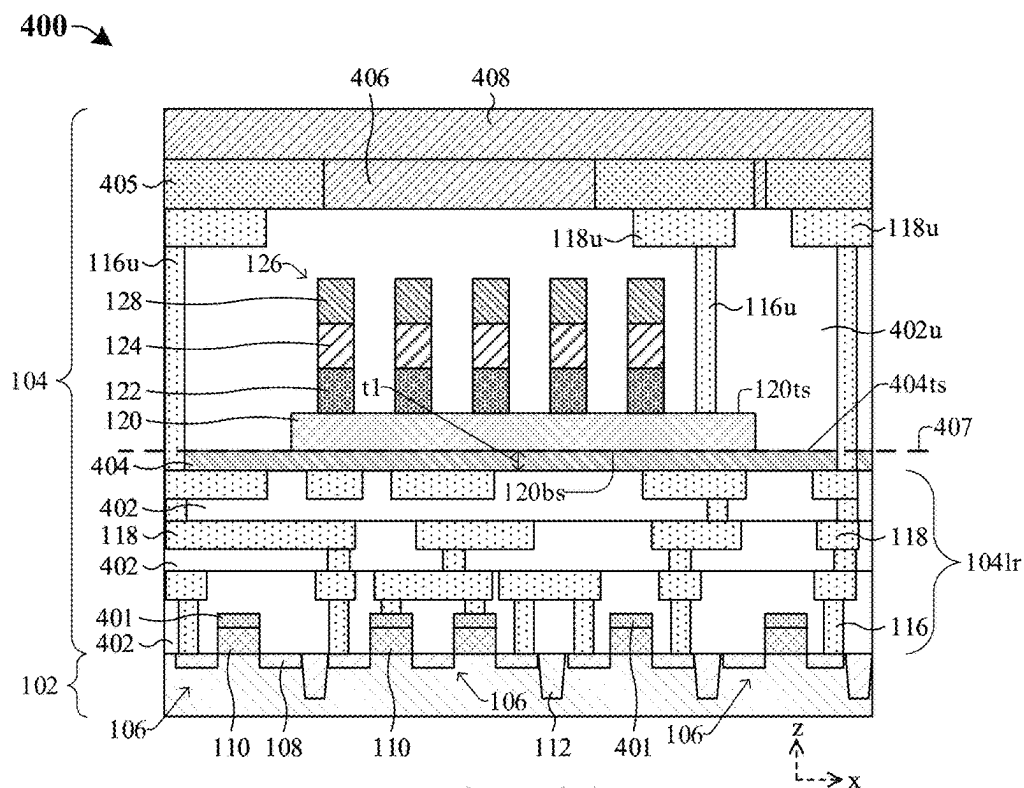


Fig. 4A

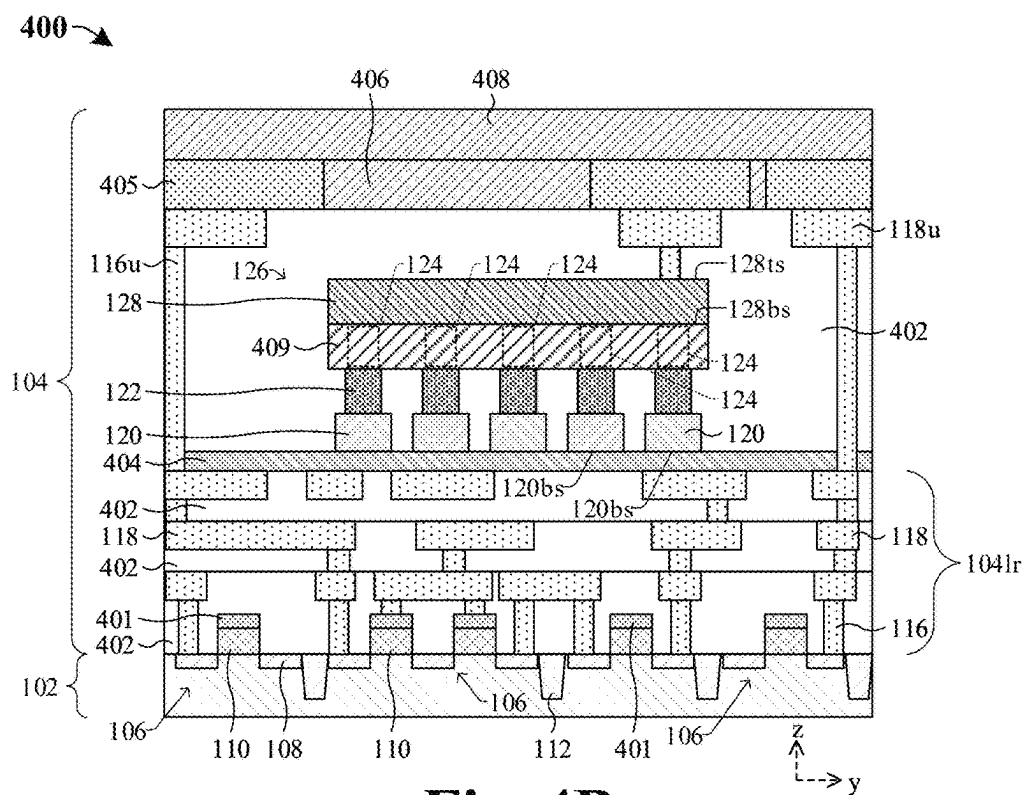


Fig. 4B

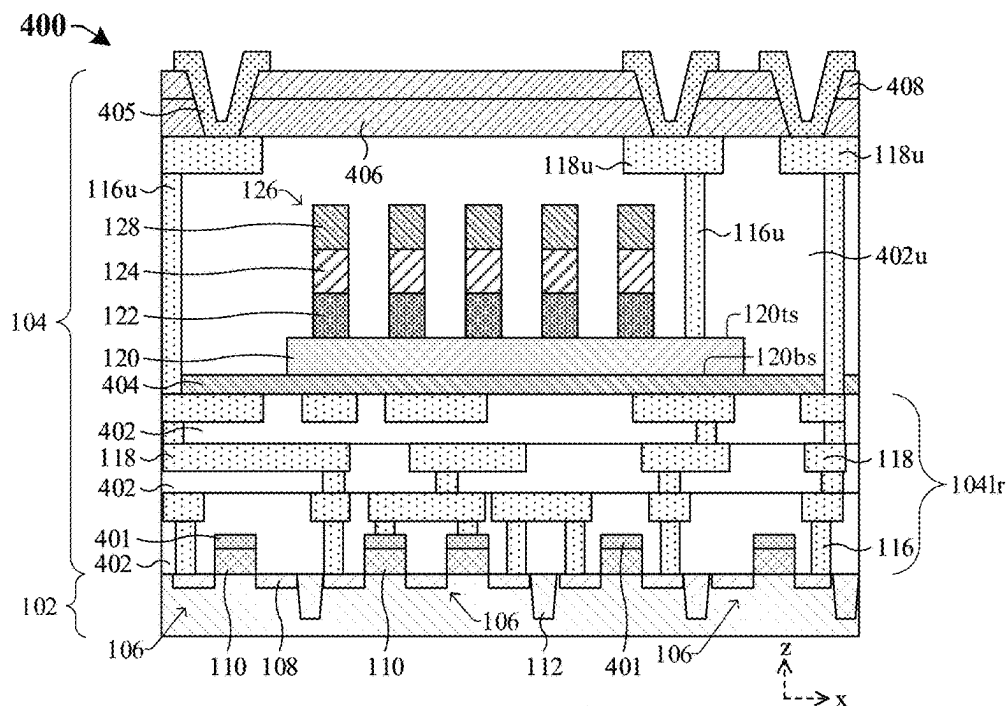


Fig. 4C

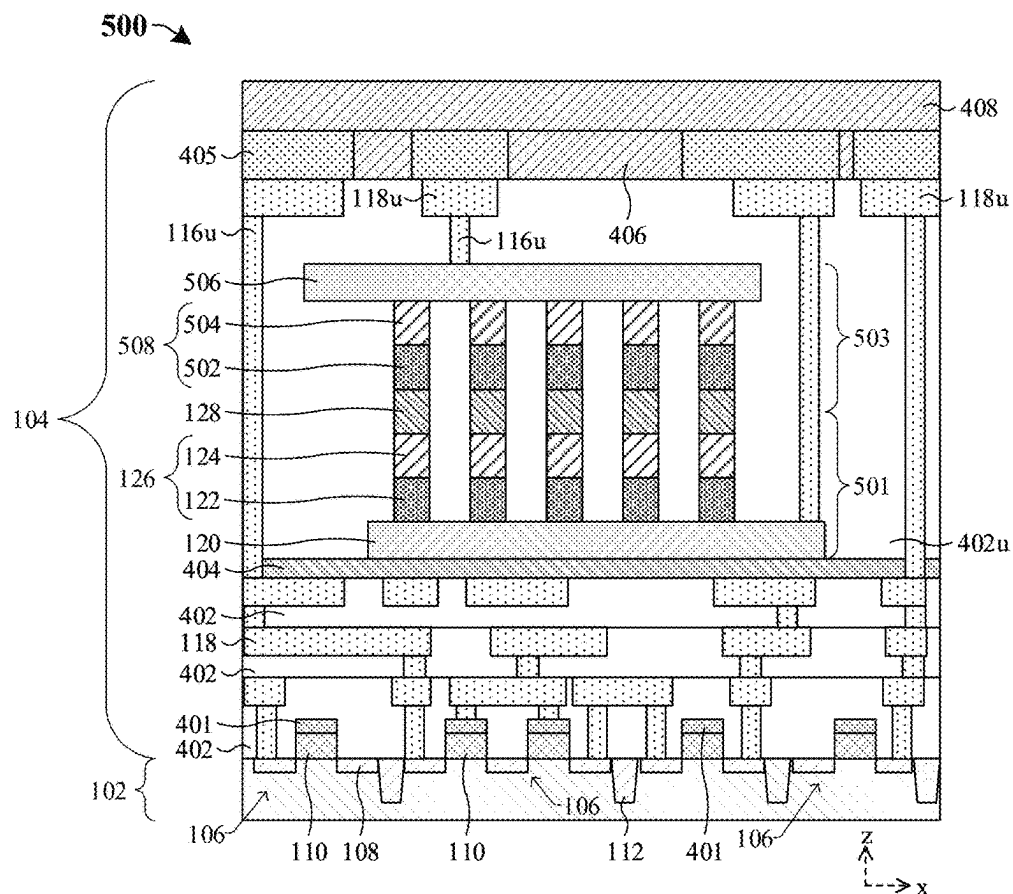


Fig. 5

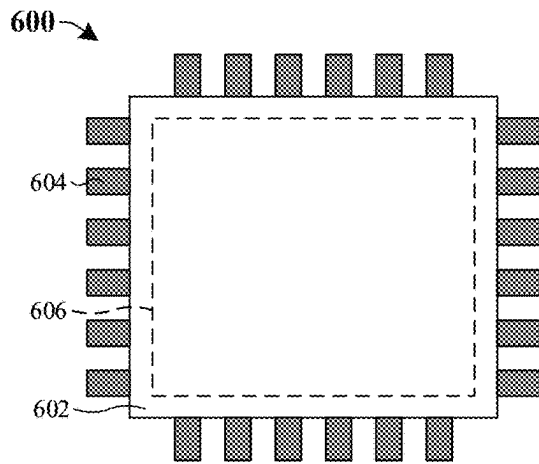


Fig. 6A

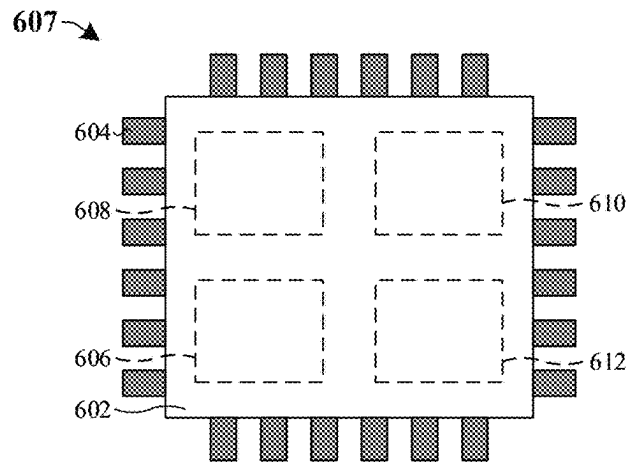


Fig. 6B

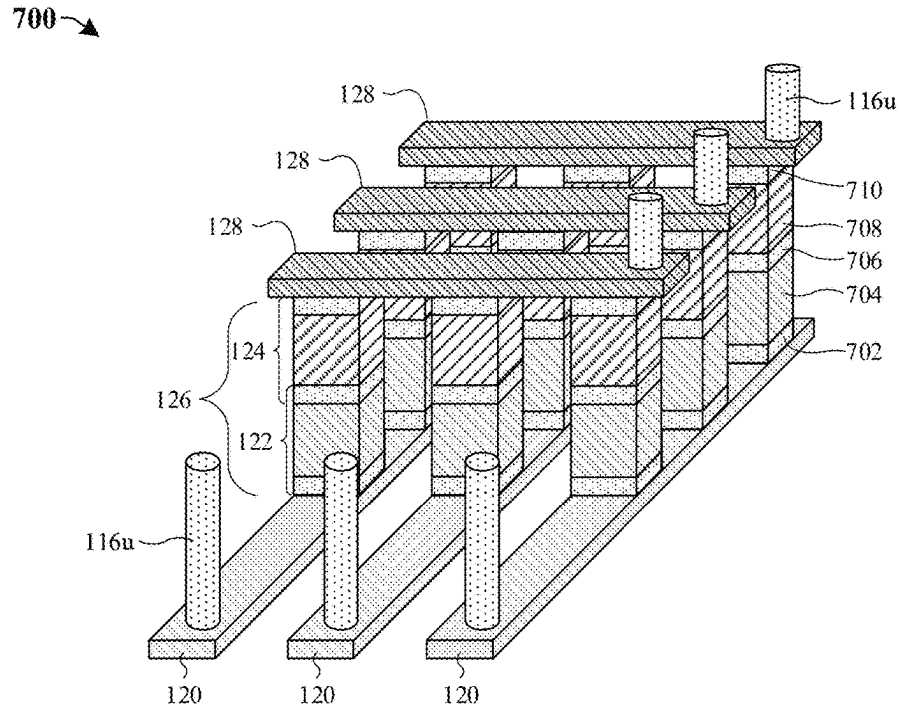


Fig. 7

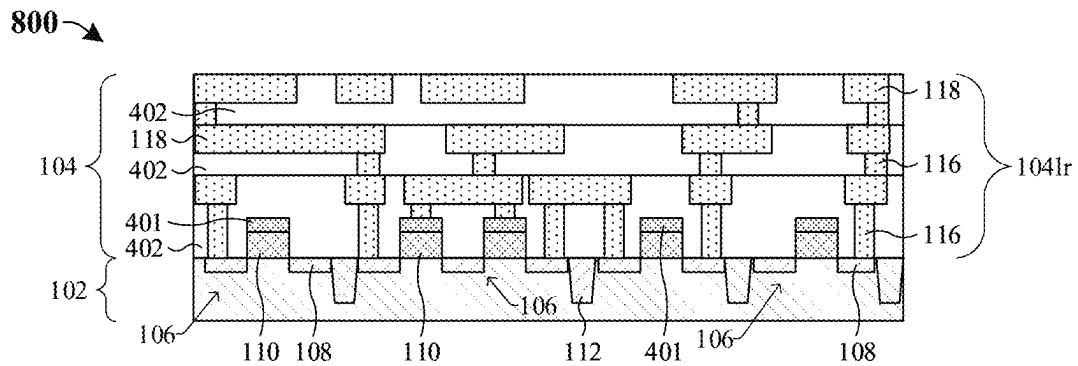


Fig. 8

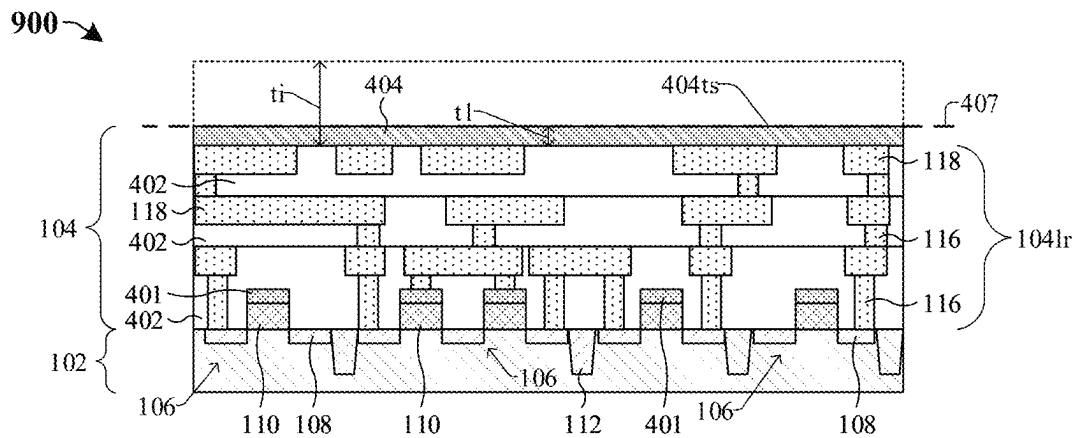


Fig. 9

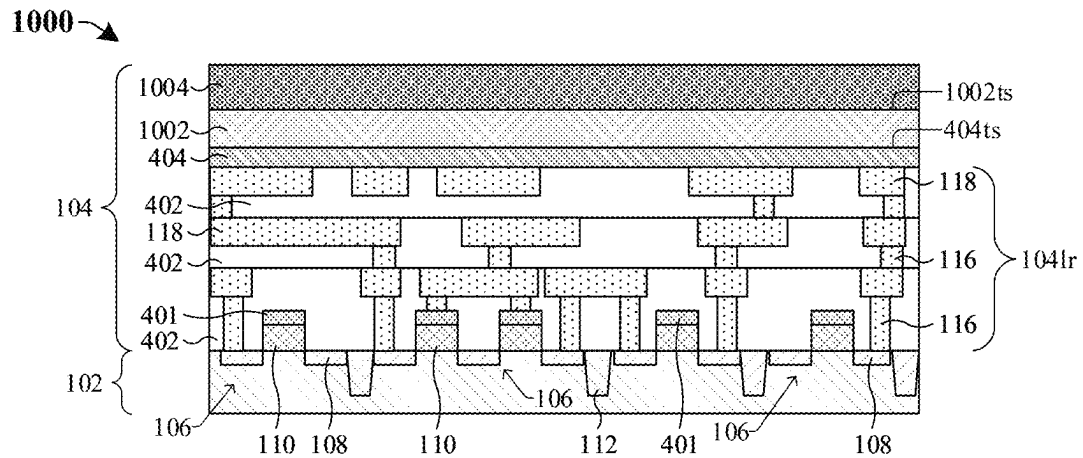


Fig. 10

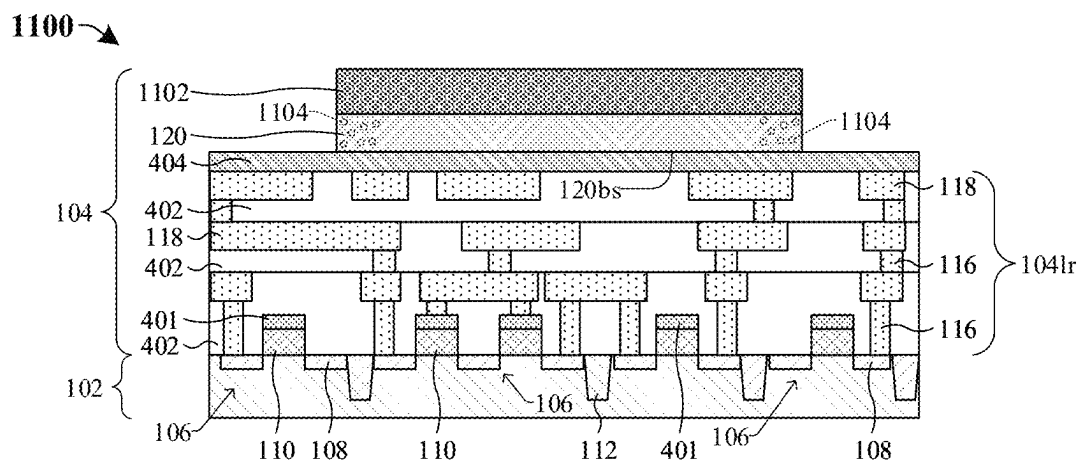


Fig. 11

1200

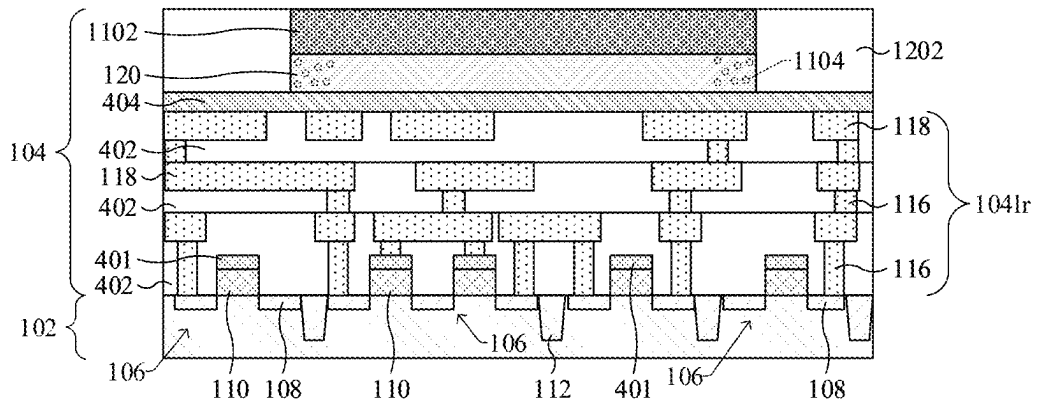


Fig. 12

1300

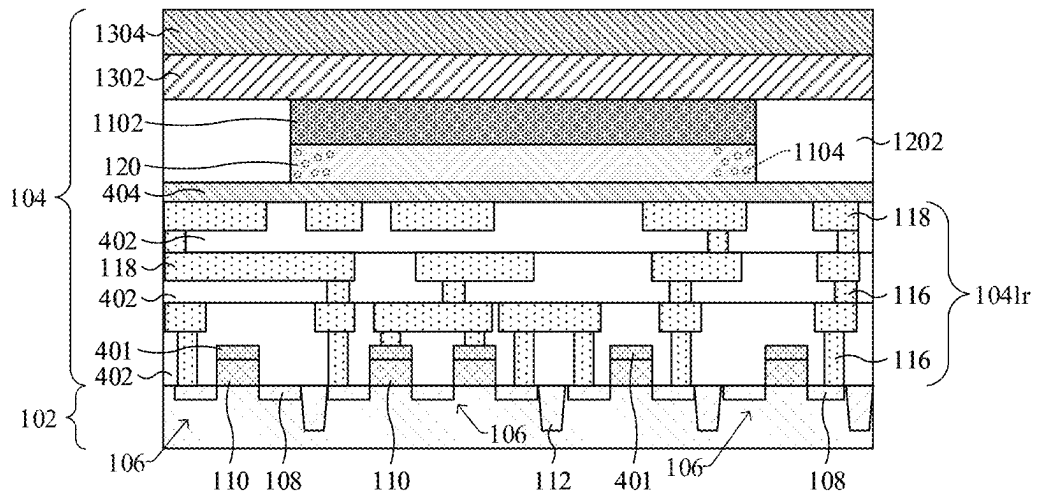


Fig. 13

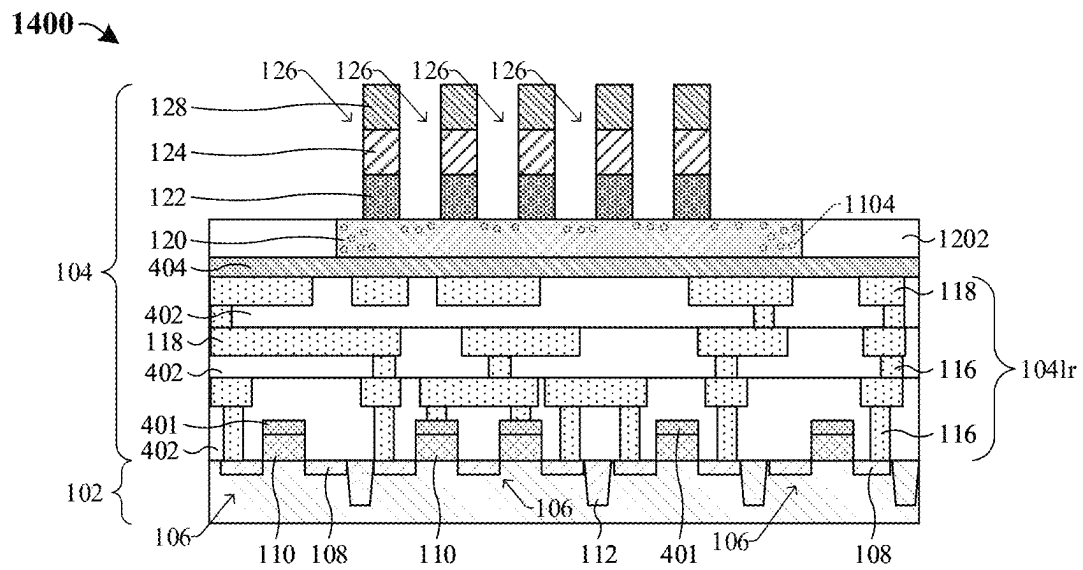


Fig. 14

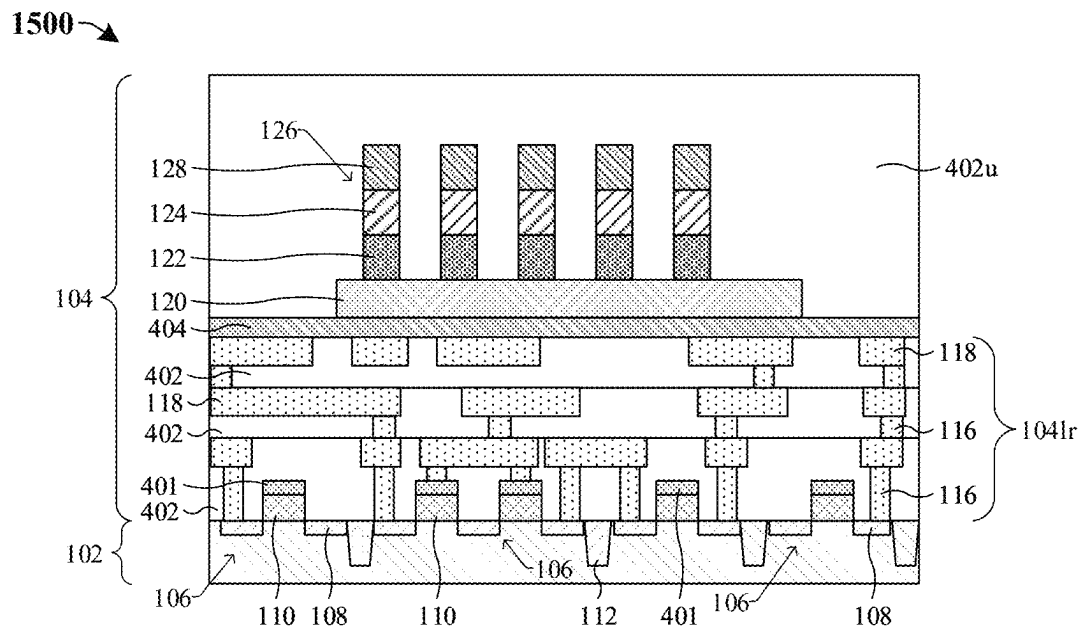
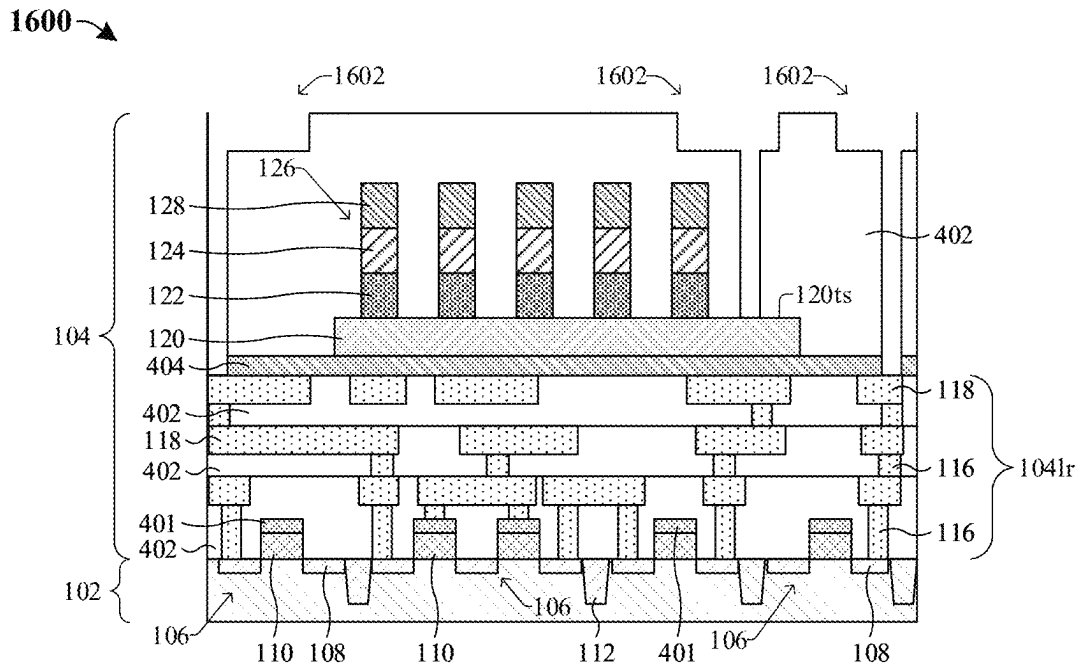
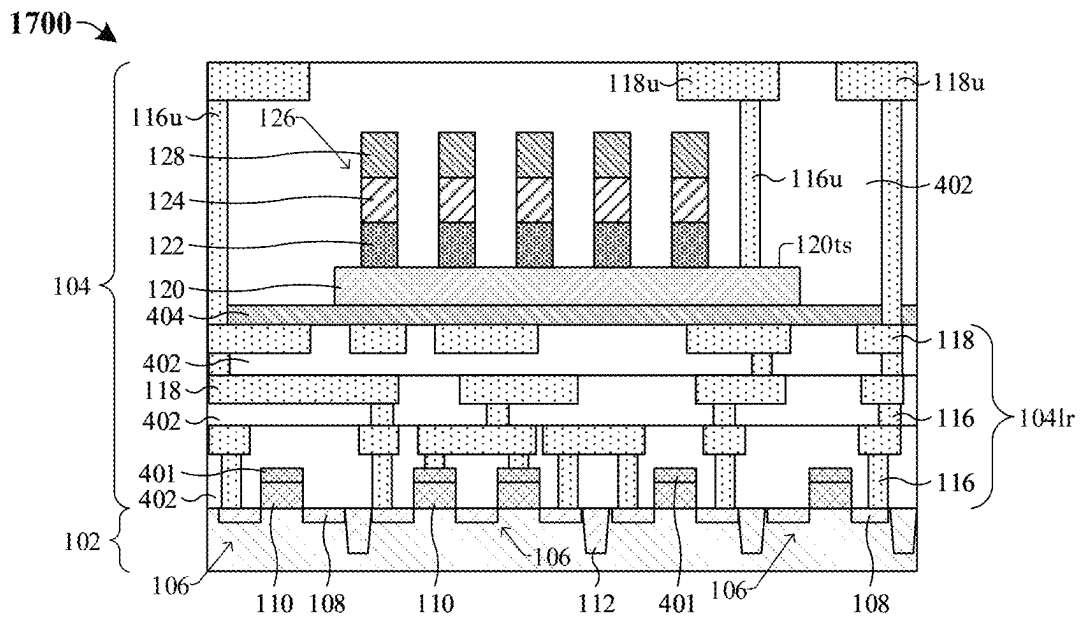


Fig. 15

**Fig. 16****Fig. 17**

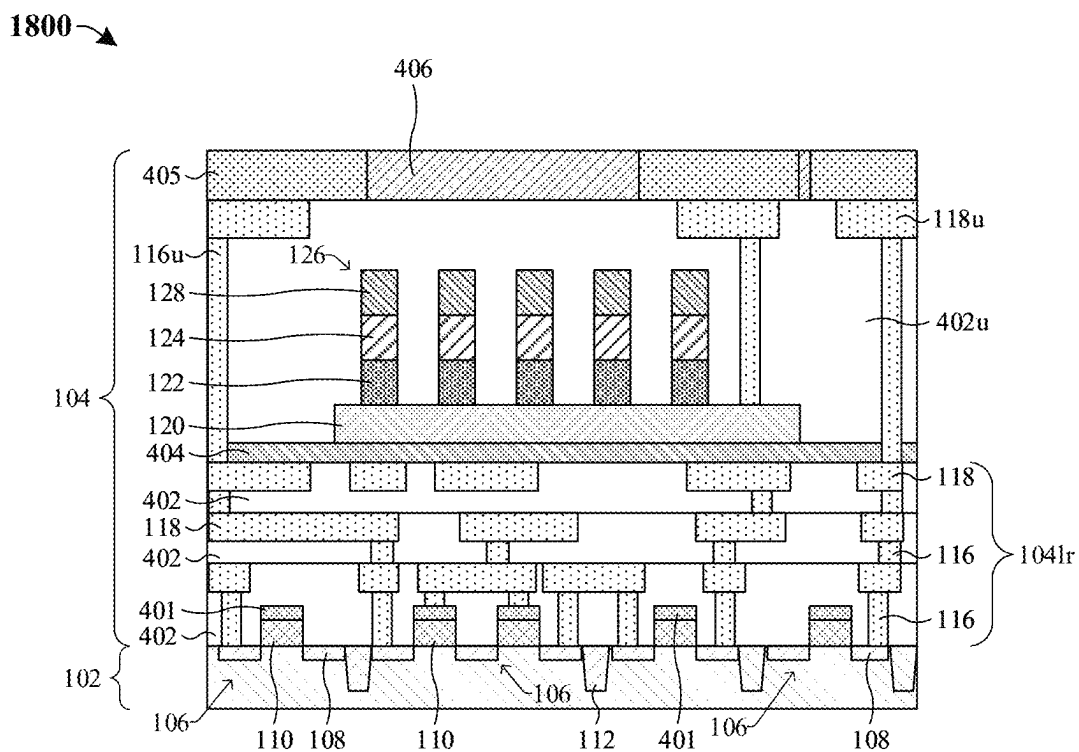


Fig. 18

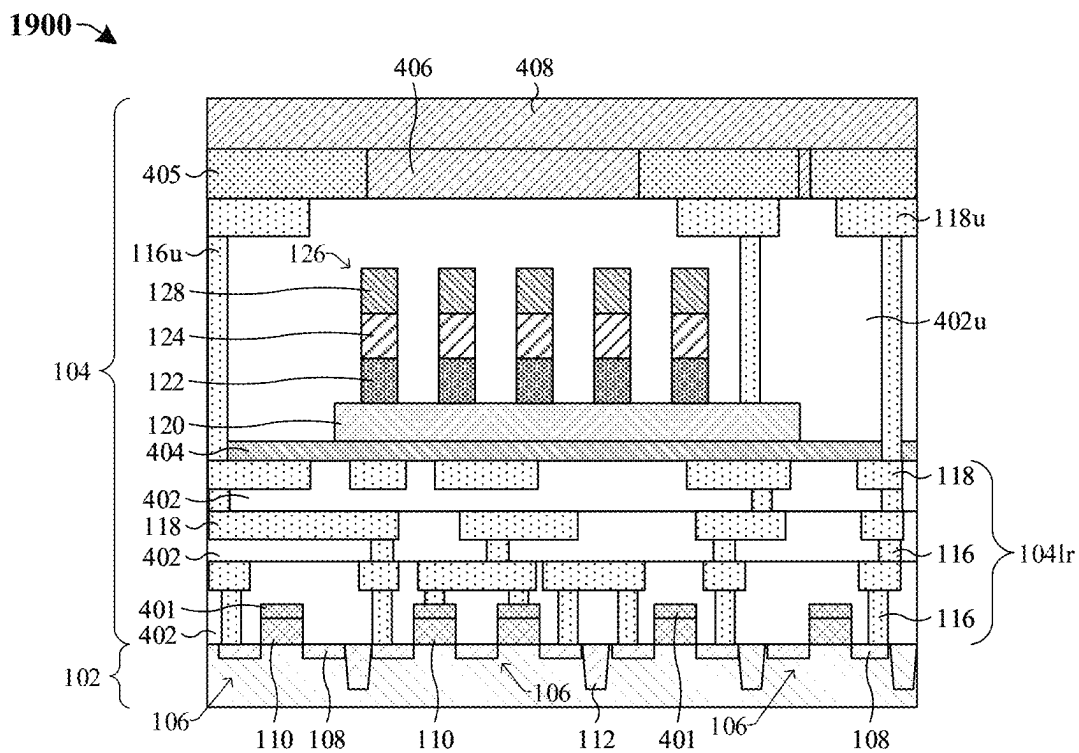
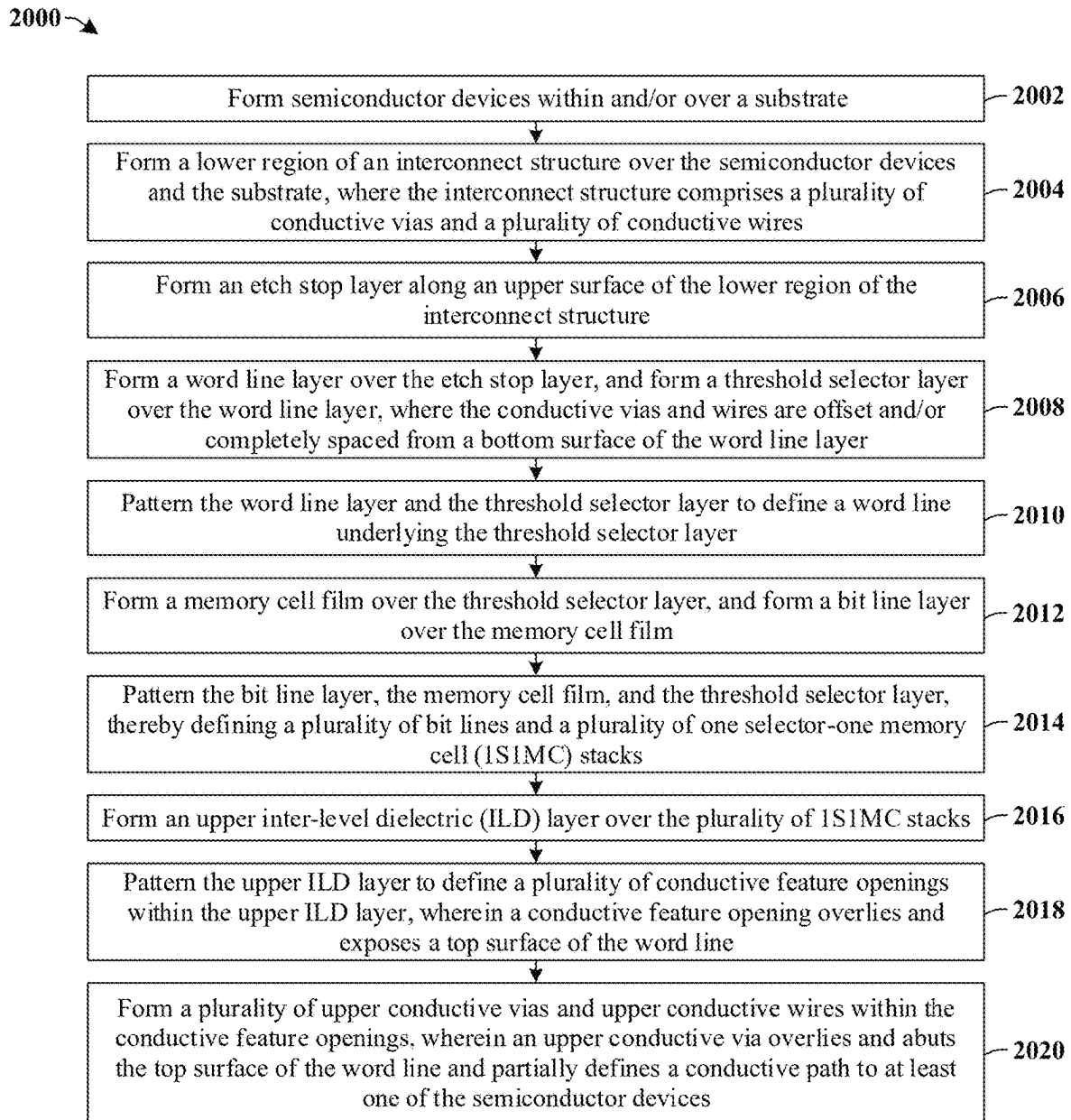


Fig. 19

**Fig. 20**

BIT LINE AND WORD LINE CONNECTION FOR MEMORY ARRAY

REFERENCE TO RELATED APPLICATIONS

This Application is a Continuation of U.S. application Ser. No. 17/555,932, filed on Dec. 20, 2021, which is a Divisional of U.S. application Ser. No. 16/821,208, filed on Mar. 17, 2020 (now U.S. Pat. No. 11,211,120, issued on Dec. 28, 2021). The contents of the above-referenced Patent Applications are hereby incorporated by reference in their entirety.

BACKGROUND

Many modern-day electronic devices include electronic memory. A cross-point memory architecture with one selector-one memory cell (1S1MC) stacks is increasingly receiving attention for use with next generation electronic memory due to its high density. Examples of next generation electronic memory include resistive random-access memory (RRAM), phase-change random-access memory (PCRAM), and magnetoresistive random-access memory (MRAM).

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIGS. 1-3 illustrate various views of some embodiments of a memory device having a plurality of memory cells disposed between overlying bit lines and underlying word lines in an interconnect structure, where an upper conductive via extends from above the memory cells to a top surface of a corresponding word line.

FIGS. 4A-C illustrate various views of some alternative embodiments of the memory device of FIGS. 1-3.

FIG. 5 illustrates a cross-sectional view of some embodiments of a three-dimensional (3D) memory array comprising a first memory array and a second memory array overlying a substrate.

FIGS. 6A-B illustrate top views of some embodiments of package structures respectively having a standalone memory region and an embedded memory region.

FIG. 7 illustrates a perspective view of some embodiments of a section of the memory device of FIGS. 1-3.

FIGS. 8-19 illustrate a series of cross-sectional views of some embodiments of a method for forming a memory device having a plurality of memory cells disposed between overlying bit lines and underlying word lines in an interconnect structure, where an upper conductive via extends from above the bit lines to a top surface of a corresponding word line.

FIG. 20 illustrates a block diagram of some embodiments of the method of FIGS. 8-19.

DETAILED DESCRIPTION

The present disclosure provides many different embodiments, or examples, for implementing different features of this disclosure. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a

first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

A cross-point memory array may include a plurality of memory cells disposed within an interconnect structure that overlies a substrate. The memory cells define a memory array and are arranged in a plurality of rows and a plurality of columns (e.g., in a crossbar array). A plurality of word lines is disposed beneath the memory array, and a plurality of bit lines is disposed over the memory array. The word lines each extend along an individual row in the memory array and are each coupled to the memory cells in the individual row. The bit lines each extend along an individual column in the memory array and are each coupled to the memory cells in the individual column. The plurality of bit lines and the plurality of word lines are electrically coupled to semiconductor devices (e.g., transistors) disposed within/over the substrate by way of conductive vias and conductive wires disposed within the interconnect structure. Thus, bias voltages may be applied to the plurality of bit lines and the plurality of word lines by way of the conductive vias and wires in the interconnect structure to select an individual memory cell (e.g., to read/write/erase the cell).

One challenge with the cross-point memory array is a connection between the word lines and corresponding lower conductive vias within the interconnect structure. For example, the interconnect structure includes lower conductive vias and lower conductive wires directly electrically coupled to a transistor disposed within and/or on the substrate. During fabrication, a first word line may be formed over a lower conductive via that is directly electrically coupled to the transistor, such that there is an electrical path between the first word line and the transistor. A process for forming the first word line may include reactive-ion etching of the conductive material of the first word line. A plasma used in the reactive-ion etching may add charge carriers (e.g., electrons) to the first word line. Because the first word line is relatively long, the first word line may act as an antenna and hence a large amount of charge carriers may build up in the first word line. The large amount of charge carriers can travel across the lower conductive via to the transistor, thereby causing breakdown in a gate dielectric layer of the transistor.

Further, the first word line may comprise a first material (e.g., tungsten) different than a second material (e.g., aluminum, copper, an alloy of the foregoing, etc.) of the lower conductive via. During fabrication, the first word line and the lower conductive via are exposed to one or more

processing fluids (e.g., etch residue cleaning fluids, deionized water, etc.). When the lower conductive via is exposed to the one or more processing fluids, galvanic corrosion of the lower conductive via may occur (e.g., due a reaction between the one or more processing fluids and the second material). This may result in formation of an oxide at the interface between the lower conductive via and the first word line, thereby causing delamination between the lower conductive via and the first word line and/or causing delamination of layers disposed over the first word line (e.g., layers making up the memory cells). Accordingly, performance of the cross-point memory array may be negatively affected (e.g., reduced yield, memory cell breakdown, transistor breakdown, etc.)

Various embodiments of the present disclosure are directed toward an integrated chip having a cross-point memory array with a plurality of memory cells disposed between a plurality of word lines and a plurality of bit lines. In various embodiments, the bit lines overlie the memory cells and the word lines underlie the memory cells. The memory array is disposed within an interconnect structure including conductive features that are disposed within a dielectric structure and that are electrically coupled to the bit lines and the word lines. The conductive features include lower conductive vias and lower conductive wires underlying the word lines, and upper conductive vias and upper conductive wires overlying the bit lines. The lower conductive vias and wires are offset from a bottom surface of each word line by a non-zero distance, such that the dielectric structure continuously extends along the bottom surface of each word line and such that the bottom surface of each word line is completely spaced from the lower conductive vias and wires. Further, an upper conductive via extends from an upper conductive wire, which is disposed above the bit lines, to contact with a top surface of each word line. This forms an electrical connection between each word line and a corresponding semiconductor device (e.g., transistor) disposed within and/or on the substrate.

During fabrication of the integrated chip, the upper conductive vias are formed after the word lines, such that the electrical paths between the word lines and the corresponding semiconductor device are formed after fabrication of the word lines. Charge carriers (e.g., electrons) accumulated in the word lines (e.g., due to a reactive-ion etch) may be dispersed before forming the upper conductive vias over the top surface of each word line. For example, before forming the upper conductive vias, a charge carrier removal process may be performed on the word lines to remove the accumulated charge carriers. This may reduce breakdown of the semiconductor devices. Further, because the lower conductive vias and wires are offset the bottom surface of each word line, galvanic corrosion between the word lines and the lower conductive vias may not occur. Accordingly, this reduces delamination of the word lines and/or memory cells, thereby increasing a performance (e.g., yield) of the cross-point memory array.

FIGS. 1-3 illustrate various views of some embodiments of a memory device **100** having a plurality of one selector-one memory cell (1S1MC) stacks **126** disposed between a plurality of word lines **120** and a plurality of bit lines **128**. FIG. 3 illustrates a top view of some embodiments of the memory device **100**. FIG. 1 illustrates a cross-sectional view of some embodiments of the memory device **100** taken along line A-A' of FIG. 3. FIG. 2 illustrates a cross-sectional view of some alternative embodiments of the memory device **100** taken along line B-B' of FIG. 3.

The memory device **100** includes an interconnect structure **104** overlying a substrate **102**. Semiconductor devices **106** are disposed within and/or over the substrate **102**. In some embodiments, the semiconductor devices **106** may be configured as transistors. In such embodiments, the semiconductor devices **106** may include a gate structure **110** and source/drain regions **108** disposed within the substrate **102** on opposing sides of the gate structure **110**. In some embodiments, the gate structure **110** includes a gate electrode overlying a gate dielectric layer. Isolation structures **112** are disposed within the substrate **102** between adjacent semiconductor devices **106**. The interconnect structure **104** includes an interconnect dielectric structure **114**, a plurality of conductive vias **116**, and a plurality of conductive wires **118**. The conductive vias **116** and the conductive wires **118** are disposed within the interconnect dielectric structure **114** and are configured to electrically couple devices disposed within the memory device **100** to one another.

The plurality of word lines **120** and the plurality of bit lines **128** are disposed within the interconnect dielectric structure **114** between upper and lower layers of the conductive vias and wires **116**, **118**. In some embodiments, the word lines **120** may be referred to as first conductive lines, and the bit lines **128** may be referred to as second conductive lines. The plurality of word lines **120** each extend laterally in a first direction (e.g., along the x-axis). In various embodiments, the word lines **120** are arranged in parallel with one another. Further, the plurality of bit lines **128** each extend laterally in a second direction (e.g., along the y-axis) transverse the first direction. In some embodiments, the bit lines **128** are arranged in parallel with one another. In further embodiments, the first direction is orthogonal to the second direction. In some embodiments, the word lines **120** and the bit lines **128** may, for example, respectively be or comprise copper, aluminum, tungsten, titanium nitride, tantalum nitride, another suitable conductive material, or any combination of the foregoing. In further embodiments, the word lines **120** and the bit lines **128** may, for example, respectively comprise a single material, such as tungsten.

The plurality of 1S1MC stacks **126** are disposed between the plurality of word lines **120** and the plurality of bit lines **128**. In various embodiments, the 1S1MC stacks **126** are arranged in an array having a plurality of rows and a plurality of columns. In some embodiments, an individual word line **120** and an individual bit line **128** are coupled to each individual 1S1MC stack **126**. In further embodiments, respective word lines **120** are coupled to respective rows of 1S1MC stacks **126**. In yet further embodiments, respective bit lines **128** are coupled to respective columns of 1S1MC stacks **126**.

Each 1S1MC stack **126** includes a memory cell **124** overlying a threshold selector **122**. The memory cell **124** is configured to store data and may be a non-volatile memory cell or a volatile memory cell. In some embodiments, the memory cell **124** may be a resistive switching memory cell (e.g., resistive random-access memory (RRAM) cell, phase-change random-access memory (PCRAM) cell, metal-cation RRAM, etc.) configured to store data based on a resistive state of a data storage structure. For example, the data storage structure may have a high resistance state associated with a first data state (e.g., binary "0") or a low resistance state associated with a second data state (e.g., binary "1"). In some embodiments, the data storage structure may comprise, for example, a chalcogenide, an oxide, a nitride, a high-k dielectric, some other suitable dielectric, or any combination of the foregoing. In further embodiments, the memory cell **124** may be magnetoresistive random-access

memory (MRAM) cell. In such an embodiment, the data storage structure may comprise a magnetic tunnel junction (MTJ) configured to store data based on a magnetic orientation of the MTJ. In addition, the threshold selector **122** is configured to switch between a low resistance state and a high resistance state depending on a voltage applied across the threshold selector **122**. For example, the threshold selector **122** may be in a high resistance state if a voltage applied across the threshold selector **122** is less than the threshold voltage, and the threshold selector **122** may have a low resistance state if a voltage across the threshold selector **122** is greater than the threshold voltage.

In some embodiments, the interconnect dielectric structure **114** continuously extends between opposing outer side-walls of each word line **120** along a bottom surface **120bs** of each word line **120**. Thus, conductive vias **116** and conductive wires **118** disposed in a lower region **104/r** of the interconnect structure **104** are offset from the bottom surface **120bs** of each word line **120**. In some embodiments, the bottom surface **120bs** does not directly contact any of the conductive vias **116** and/or is completely spaced from the conductive vias **116**. In some embodiments, the bottom surface **120bs** directly contacts the interconnect dielectric structure **114** throughout an entirety of the bottom surface **120bs**. Further, upper conductive wires **118u** and upper conductive vias **116u** overlie the plurality of bit lines **128** and word lines **120**. Upper conductive vias **116u** continuously extend from upper conductive wires **118u** to top surfaces **120ts** of respective word lines **120**. Therefore, the word lines **120** may be electrically coupled to respective semiconductor devices **106** by way of the upper conductive vias **116u**.

In some embodiments, during fabrication of the memory device **100**, a process for forming the word lines **120** may include: depositing a conductive material (e.g., tungsten) over the lower region **104/r** of the interconnect structure **104**; and subsequently performing a dry etch on the conductive material to define the word lines **120**. The dry etch may include exposing the conductive material to a plasma, such that charge carriers (e.g., electrons) accumulate in the word lines **120**. Because the upper conductive vias **116u** are formed after forming the word lines **120**, charge carriers accumulated in the word lines **120** may be dispersed before forming the electrical paths between the word lines **120** and respective semiconductor devices **106**. This may reduce the likelihood of breakdown of the gate dielectric layers in the semiconductor devices **106**. In addition, in some embodiments, the conductive vias and wires **116**, **118** within the lower region **104/r** of the interconnect structure **104** are offset and/or completely spaced from the bottom surface **120bs** of each word line **120**. This prevents galvanic corrosion between the word lines **120** and the conductive vias and wires **116**, **118** within the lower region **104/r** of the interconnect structure **104**. Accordingly, delamination of the word lines **120**, the 1S1MC stacks **126**, and/or the bit lines **128** is reduced, thereby increasing a performance of the memory device **100**.

As illustrated in the top view of FIG. 3, the word lines **120** extend continuously along the first direction (e.g., the x-axis) and are parallel to one another. Further, the bit lines **128** extend along the second direction (e.g., the y-axis) and are parallel to one another. In some embodiments, the first direction is orthogonal to the second direction. In addition, one or more upper conductive vias **116u** directly overlie(s) corresponding word lines **120** and corresponding bit lines **128**.

FIG. 4A illustrates a cross-sectional view of some embodiments of an integrated chip **400** including a plurality of one selector-one memory cell (1S1MC) stacks **126** disposed between a plurality of word lines **120** and a plurality of bit lines **128**. In some embodiments, the cross-sectional view of FIG. 4A may be taken along the x-z plane. In further embodiments, FIG. 4A illustrates a cross-sectional view of some alternative of the memory device **100** taken along line A-A' of FIG. 3.

The integrated chip **400** includes an interconnect structure **104** overlying a substrate **102**. In some embodiments, the substrate **102** may, for example, be or comprise monocrystalline silicon/CMOS bulk, silicon-germanium (SiGe), silicon on insulator (SOI), or another suitable substrate material and/or may comprise a first doping type (e.g., p-type). The interconnect structure **104** includes a plurality of conductive vias **116**, a plurality of conductive wires **118**, and an interconnect dielectric structure. In some embodiments, the interconnect dielectric structure includes a plurality of inter-level dielectric (ILD) layers **402**, an etch stop layer **404**, and passivation layers **406**, **408**. In some embodiments, the plurality of conductive vias and wires **116**, **118** may, for example, respectively be or comprise aluminum, copper, titanium nitride, tantalum nitride, another suitable conductive material, or any combination of the foregoing. In further embodiments, the plurality of ILD layers **402** may, for example, respectively be or comprise an oxide, such as silicon dioxide, a low-k dielectric material, another suitable dielectric material, or any combination of the foregoing. In yet further embodiments, the etch stop layer **404** may, for example, be or comprise silicon dioxide, a low-k dielectric material, silicon nitride, silicon carbide, extreme low k (ELK) dielectric material, another suitable dielectric material, or any combination of the foregoing. In various embodiments, the passivation layers **406**, **408** may, for example, respectively be or comprise silicon dioxide, silicon oxynitride, silicon oxycarbide, silicon nitride, silicon carbide, another suitable dielectric material, or any combination of the foregoing.

A plurality of semiconductor devices **106** are disposed within and/or over the substrate **102**. In some embodiments, the semiconductor devices **106** may, for example, be configured as transistors or as another suitable semiconductor device. In such embodiments, the semiconductor devices **106** may include corresponding source/drain regions **108**, corresponding gate structures **110**, and corresponding gate capping layers **401**. In some embodiments, the source/drain regions **108** are disposed within the substrate **102** and may comprise a second doping type (e.g., n-type) opposite the first doping type (e.g., p-type). In further embodiments, the gate structures **110** may include corresponding gate electrodes overlying corresponding gate dielectric layer. In various embodiments, the gate electrodes may, for example, be or comprise a metal (such as aluminum, tungsten, titanium, any combination of the foregoing, or the like), polysilicon, another suitable conductive material, or any combination of the foregoing. In further embodiments, the gate dielectric layers may, for example, be or comprise silicon dioxide, a high-k dielectric material, another suitable dielectric material, or any combination of the foregoing. The gate capping layers **401** are conductive and may, for example, be or comprise tantalum, titanium, a silicide, another suitable material, or any combination of the foregoing. Further, isolation structures **112** are disposed within the substrate **102** and may laterally surround a corresponding semiconductor device **106**. In some embodiments, the isolation structures **112** may, for example, be configured as shallow trench

isolation (STI) structures, deep trench isolation (DTI) structures, or another suitable isolation structure. In further embodiments, the isolation structures **112** may, for example, be or comprise silicon dioxide, silicon nitride, silicon carbide, another suitable dielectric material, or any combination of the foregoing.

The plurality of word lines **120** and the plurality of bit lines **128** are disposed within an upper ILD layer **402u** overlying a lower region **104lr** of the interconnect structure **104**. In some embodiments, the etch stop layer **404** continuously extends along a top surface of a topmost ILD layer **402** in the lower region **104lr**. A top surface **404ts** of the etch stop layer **404** continuously and laterally extends along a bottom surface **120bs** of each word line **120**. In some embodiments, the top surface **404ts** of the etch stop layer **404** continuously extends along the bottom surface **120bs** of each word line **120** along an unbroken path and/or directly contacts an entirety of the bottom surface **120bs** of each word line **120**. In yet further embodiments, the etch stop layer **404** is configured to separate (e.g., completely separate) the word lines **120** from conductive vias **116** and conductive wires **118** disposed within the lower region **104lr** of the interconnect structure **104**. In various embodiments, the conductive vias **116** and the conductive wires **118** disposed within the lower region **104lr** of the interconnect structure **104** are offset from the bottom surface **120bs** of each word line **120** by a non-zero distance.

In further embodiments, the etch stop layer **404** may, for example, be or comprise an oxide, such as silicon dioxide, a low-k dielectric material, another dielectric material, or any combination of the foregoing and/or may have a thickness **t1** within a range of about 100 to 1,000 Angstroms or less than about 1,000 Angstroms. It will be appreciated that other values for the thickness **t1** are within the scope of the disclosure. In some embodiments, the top surface **404ts** of the etch stop layer **404** is substantially flat (e.g., a flat top surface within a tolerance of a chemical mechanical planarization (CMP) process). For example, in some embodiments, at any point a height of the top surface **404ts** of the etch stop layer **404** varies within a range of about -5% to +5% of the thickness **t1** from a level horizontal line **407** located between the top surface **404ts** of the etch stop layer **404** and the bottom surface **120bs** of the word line **120**. For example, if the thickness **t1** is about 1,000 Angstroms, then the height of the top surface **404ts** of the etch stop layer **404** varies within a range of about -50 Angstroms to +50 Angstroms from the level horizontal line **407**. In other embodiments, at any point the height of the top surface **404ts** of the etch stop layer **404** varies within a range of about -5 Angstroms to +5 Angstroms from the level horizontal line **407**.

In some embodiments, during fabrication, the word line **120** is deposited along the top surface **404ts** of the etch stop layer **404**. Thus, the bottom surface **120bs** of the word line **120** may be substantially flat due to the top surface **404ts** of the etch stop layer **404** being substantially flat. In some embodiments, the word line **120** comprises a single material, such as tungsten. The single material may have a relatively high hardness, such that the substantially flat top surface **404ts** of the etch stop layer **404** prevents stress in the word line **120**. This, in part, mitigates delamination of the word line **120** and structures overlying the word line **120** (e.g., the bit lines **128** and/or the 1S1MC stacks **126**).

The plurality of 1S1MC stacks **126** are disposed between the plurality of word lines **120** and the plurality of bit lines **128**, such that the 1S1MC stacks **126** overlie top surfaces **120ts** of corresponding word lines **120**. In various embodiments, the 1S1MC stacks **126** are arranged in an array

having a plurality of rows and a plurality of columns. In some embodiments, an individual word line **120** and an individual bit line **128** are coupled to each individual 1S1MC stack **126**. In further embodiments, respective word lines **120** are coupled to respective rows of 1S1MC stacks **126**. In yet further embodiments, respective bit lines **128** are coupled to respective columns of 1S1MC stacks **126**. Each 1S1MC stack **126** includes a memory cell **124** overlying a threshold selector **122**. The memory cell **124** is configured to store data and may be a non-volatile memory cell or a volatile memory cell.

Upper conductive wires **118u** and upper conductive vias **116u** are disposed within the upper ILD layer **402u**. In some embodiments, the upper conductive wires **118u** are disposed vertically above the bit lines **128**. The upper conductive vias **116u** extend from the upper conductive wires **118u** to the word lines **120**, the bit lines **128**, and/or conductive wires **118** disposed within the lower region **104lr** of the interconnect structure **104**. Thus, the upper conductive vias **116u** are configured to electrically couple the word lines **120** and/or the bit lines **128** to the semiconductor devices **106** and/or another integrated chip (not shown). By virtue of the upper conductive vias **116u** contacting the top surfaces **120ts** of the word lines **120** instead of contacting the bottom surfaces **120bs** of the word lines **120**, damage to the semiconductor devices **106** (e.g., due to accumulated charge carriers in the word lines **120**) and/or the word lines **120** (e.g., due to delamination of the word lines **120**) may be reduced. This, in turn, increases a performance of the integrated chip **400**.

The passivation layers **406**, **408** overlie the upper ILD layer **402u** and the upper conductive vias and wires **116u**, **118u**. Further, bond pads **405** are disposed within a first passivation layer **406** and overlie corresponding upper conductive wires **118u**. In some embodiments, the bond pads **405** may extend through the second passivation layer **408** and/or may comprise an opening out of view (e.g., see FIG. 4C). Furthermore, the second passivation layer **408** overlies the first passivation layer. The bond pads **405** may be configured to electrically couple the conductive wires and vias **116**, **118** to other semiconductor devices disposed on another integrated chip (not shown).

FIG. 4B illustrates a cross-sectional view of some alternative embodiments of the integrated chip **400** of FIG. 4A. In some embodiments, the cross-sectional view of FIG. 4B may be taken along the y-z plane, which is orthogonal to the x-z plane. In further embodiments, FIG. 4B illustrates a cross-sectional view of some alternative of the memory device **100** taken along line B-B' of FIG. 3.

As illustrated in the cross-sectional view of FIG. 4B, in some embodiments, a memory cell layer **409** is disposed along a bottom surface **128bs** of each bit line **128**. In such embodiments, the memory cell layer **409** is disposed between a bit line **128** and a corresponding column of threshold selectors **122**. In further embodiments, a memory cell **124** is defined in a region of the memory cell layer **409** disposed between the bit line **128** and a corresponding threshold selector **122**. In yet further embodiments, the memory cell layer **409** includes a data storage structure that is disposed between the bit line **128** and the threshold selectors **122**. In some embodiments, the data storage structure may comprise, for example, a chalcogenide, an oxide, a nitride, a high-k dielectric, or some other suitable dielectric.

For example, in some embodiments, the memory cells **124** may each be configured as an RRAM cell. In such embodiments, the memory cell layer **409** has a data storage structure that may, for example, be or comprise gold and/or

hafnium oxide, copper and hafnium oxide, aluminum and hafnium oxide, arsenic and hafnium oxide, gold tellurium and hafnium oxide, silicon oxide, titanium oxide, aluminum oxide (e.g., Al_2O_3), tantalum oxide, zirconium oxide, or another suitable material. In such embodiments, by applying appropriate bias conditions to the bit line 128 and the word lines 120, each memory cell 124 may be switched between a first state with a low resistance and a second state with a high resistance. In further embodiments, in the first state, a conductive filament may be made in the data storage structure of the memory cell layer 409 between the bit line 128 and a corresponding threshold selector 122 disposed below the memory cell layer 409. In some embodiment, the conductive filament may be confined within the dashed box illustrating the location of the corresponding memory cell 124, thereby ensuring data states of adjacent memory cells 124 are isolated from one another. In yet further embodiments, in the second state, at least a portion of the conductive filament is unmade in the memory cell layer 409 between the bit line 128 and the corresponding threshold selector 122. Each memory cell 124 may be individually switched between the first and second states as described above.

FIG. 4C illustrates a cross-sectional view of some alternative embodiments of the integrated chip 400 of FIG. 4A, in which the bond pads 405 are disposed within the first and second passivation layers 406, 408. In some embodiments, the bond pads 405 may, for example comprise aluminum and/or may be electrically coupled to another integrated chip by contact wires (not shown). It will be appreciated that the bond pads 405 comprising another material is also within the scope of the disclosure.

FIG. 5 illustrates a cross-sectional view of some alternative embodiments of a three-dimensional (3D) memory array 500 comprising a first memory array 501 and a second memory array 503 overlying the substrate 102.

The first and second memory arrays 501, 503 are stacked, such that the second memory array 503 overlies and is spaced from the first memory array 501. In some embodiments, the first memory array 501 comprises a plurality of one selector-one memory cell (1S1MC) stacks 126 disposed between the plurality of word lines 120 and the plurality of bit lines 128. The 1S1MC stacks 126 each comprise a memory cell 124 overlying a threshold selector 122. Further, the second memory array 503 comprises a plurality of upper 1S1MC stacks 508 disposed between the plurality of bit lines 128 and a plurality of upper word lines 506. In some embodiments, the plurality of upper word lines 506 extend in parallel with the word lines 120 and/or comprise a same material as the word lines 120. In some embodiments, the plurality of upper 1S1MC stacks 508 comprise corresponding upper memory cells 504 overlying corresponding upper thresholds selector 502. In some embodiments, the upper memory cells 504 may be configured as the memory cells 124 of FIG. 1, 2, 4A, 4B, or 4C. In further embodiments, the upper threshold selectors 502 may be configured as the threshold selectors 122 of FIG. 1, 2, 4A, 4B, or 4C.

FIG. 6A illustrates a top view of some embodiments of a package structure 600 including a plurality of electric connectors 604 extending from an integrated chip 602. In some embodiments, the integrated chip 602 may be configured as the memory device 100 of FIGS. 1-3. In such embodiments, the plurality of 1S1MC stacks (126 of FIGS. 1-3) are laterally disposed across the memory region 606. In further embodiments, the memory region 606 continuously extends across a majority of an area of the integrated chip 602. In yet further embodiments, the electric connectors 604 are con-

figured to electrically couple devices disposed within the memory region 606 to another integrated chip (not shown). In some embodiments, the memory region 606 is the only type of semiconductor region on the integrated chip 602 and/or the memory region may also be known as a stand-alone memory region.

FIG. 6B illustrates a top view of some embodiments of a package structure 607 including a plurality of electric connectors 604 extending from an integrated chip 602. In some embodiments, the integrated chip 602 includes a memory region 606, a central processing unit (CPU) region 608, a static random-access memory (SRAM) region 610, and an analog region 612 that are disposed laterally adjacent to one another over a single substrate (e.g., substrate 102 of FIGS. 1 and 2). In such embodiments, the memory region 606 may also be known as an embedded memory region. In some embodiments, the plurality of 1S1MC stacks (126 of FIGS. 1-3), the word lines (120 of FIGS. 1-3), and the bit lines (128 of FIGS. 1-3) are disposed laterally within the embedded memory region 606. Further, CPU devices (not shown) are disposed laterally within the CPU region 608, SRAM devices (not shown) are disposed laterally within the SRAM region 610, and analog devices (not shown) are disposed laterally within the analog region 612. In such embodiments, the conductive vias (116 of FIGS. 1-3) and the conductive wires (118 of FIGS. 1-3) are configured to electrically couple devices disposed within the memory region 606, the CPU region 608, the SRAM region 610, and the analog region 612 to one another. Further, the electric connectors 604 may be electrically coupled to the conductive vias (116 of FIGS. 1-3) and the conductive wires (118 of FIGS. 1-3).

FIG. 7 illustrates a perspective view of some embodiments of a memory device 700 including a plurality of 1S1MC stacks 126 disposed between a plurality of bit lines 128 and a plurality of word lines 120.

In some embodiments, the 1S1MC stacks 126 include corresponding memory cells 124 overlying corresponding threshold selectors 122. In some embodiments, a 1S1MC stack 126 includes a lower electrode 702, a middle electrode 706, a threshold selector layer 704 disposed between the lower and middle electrodes 702, 706, an upper electrode 710, and a data storage structure 708 disposed between the middle and upper electrodes 706, 710. The lower electrode 702, the middle electrode 706, and the threshold selector layer 704 define a threshold selector 122. The middle electrode 706, the upper electrode 710, and the data storage structure 708 define a memory cell 124. In some embodiments, the upper, middle, and lower electrodes 702, 706, 710 may, for example, respectively be or comprise tungsten, titanium, tantalum, another conductive material, or any combination of the foregoing.

The 1S1MC stacks 126 are arranged within a memory array comprising rows and/or columns. 1S1MC stacks 126 within a row of the memory array are operably coupled to a word line 120, while 1S1MC stacks 126 within a column of the memory array are operably coupled to a bit line 128. This causes the plurality of 1S1MC stacks 126 to be respectively associated with an address defined by an intersection of a word line and a bit line. In some embodiments, the memory array is coupled to support circuitry that is configured to read from and/or write to the plurality of 1S1MC stacks 126 by way of the upper conductive vias 116u. In some embodiments, the support circuitry comprises a bit line (BL) decoder (not shown), a control unit (not shown), a word line (WL) decoder (not shown), and/or access devices (not shown). In some embodiments, the control unit is a micro-

processor unit. In further embodiments, the access devices may be the semiconductor devices (e.g., **106** of FIGS. **1-2**).

In some embodiments, during operation of the memory device **700**, the control unit may supply an address to the WL decoder and/or the BL decoder. The address is associated with a single 1S1MC stack **126** within the memory array. The WL decoder is configured to selectively apply a signal (e.g., a current and/or voltage) to one or more of the word lines **120** based upon the received address. In addition, the BL decoder is configured to selectively apply a signal (e.g., a current and/or voltage) to one or more of the bit lines **128** based upon the received address. For example, during a read operation of the memory device **700**, the BL decoder is configured to apply a read voltage to one of the plurality of bit lines **128**, such that an output of at least one 1S1MC stack **126** may be accessed (e.g., at a source line). In further embodiments, during a write operation, the WL decoder is configured to apply a write voltage to at least one of the plurality of word lines **120** such that a resistive value of at least one 1S1MC stack **126** may be set and/or changed.

FIGS. **8-19** illustrate cross-sectional views **800-1900** of some embodiments of a method for forming a memory device having a plurality of memory cells disposed between overlying bit lines and underlying word lines in an interconnect structure, where an upper conductive via extends from above the bit lines to a top surface of a corresponding word line. Although the cross-sectional views **800-1900** shown in FIGS. **8-19** are described with reference to a method, it will be appreciated that the structures shown in FIGS. **8-19** are not limited to the method but rather may stand alone separate from the method. Although FIGS. **8-19** are described as a series of acts, it will be appreciated that these acts are not limiting in that the order of the acts can be altered in other embodiments, and the methods disclosed are also applicable to other structures. In other embodiments, some acts that are illustrated and/or described may be omitted in whole or in part.

As shown in cross-sectional view **800** of FIG. **8**, a substrate **102** is provided and isolation structures **112** are formed within the substrate **102**. In some embodiments, the substrate **102** may, for example, be a bulk substrate (e.g., a bulk silicon substrate), a silicon on insulator (SOI) substrate, or some other suitable substrate and/or may comprise a first doping type (e.g., p-type). In some embodiments, a process for forming the isolation structures **112** may include: 1) selectively etching the substrate **102** to form trenches in the substrate **102**; and 2) filling (e.g., by chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), thermal oxidation, etc.) the trenches with a dielectric material (e.g., silicon dioxide, silicon nitride, silicon carbide, etc.). In further embodiments, the substrate **102** is selectively etched by forming a masking layer (not shown) over the substrate **102**, and subsequently exposing the substrate **102** to an etchant configured to selectively remove unmasked portions of the substrate **102**.

Also shown in FIG. **8**, semiconductor devices **106** are formed over and/or within the substrate **102**. In some embodiments, the semiconductor devices **106** are configured as transistors. In such embodiments, a process for forming the semiconductor devices **106** may include deposition and/or growing (e.g., by CVD, PVD, ALD, thermal oxidation, etc.) a gate dielectric layer on a top surface of the substrate **102**. Next, a gate electrode layer may be deposited on the gate dielectric layer. Subsequently, the gate dielectric and electrode layers are patterned (e.g., by a photolithography/etching process) to form gate dielectrics and gate electrodes, respectively, thereby defining the gate structure **110**.

In further embodiments, a gate capping layer **401** is formed and/or grown over the gate electrode of the gate structure **110**. Further, source/drain regions **108** are formed on opposing sides of the gate structures **110**, thereby defining the semiconductor devices **106**. In some embodiments, the source/drain regions **108** may be formed by a selective ion implantation that utilizes a masking layer (not shown) disposed on the top surface of the substrate **102** to selectively implant dopants of a second doping type (e.g., n-type dopants) into the substrate **102**. In some embodiments, the second doping type is opposite the first doping type. In some embodiments, the gate electrode layer may comprise, for example, polysilicon, aluminum, titanium, another suitable conductive material, or any combination of the foregoing. In further embodiments, the gate dielectric layer may comprise, for example, silicon dioxide, another suitable oxide, a high-k dielectric material, another suitable dielectric material, or any combination of the foregoing. In yet further embodiments, the gate capping layer **401** may comprise, for example, tantalum, titanium, titanium nitride, tantalum nitride, another suitable material, or any combination of the foregoing.

In addition, as shown in FIG. **8**, a lower region **104/r** of an interconnect structure **104** is formed over the substrate **102**. The lower region **104/r** of the interconnect structure **104** includes a plurality of inter-level dielectric (ILD) layers **402**, a plurality of conductive vias **116**, and a plurality of conductive wires **118**. In some embodiments, the ILD layers **402** may, for example, be deposited by CVD, PVD, ALD, or another suitable growth or deposition process. Further, the ILD layers **402** may, for example, be or comprise silicon dioxide, a low-k dielectric material, another suitable material, or any combination of the foregoing. Further, in some embodiments, the plurality of conductive vias **116** and the plurality of conductive wires **118** may be formed by a single damascene process or a dual damascene process. In further embodiments, the plurality of conductive vias and/or wires **116**, **118** may, for example, respectively be or comprise copper, aluminum, titanium nitride, tantalum nitride, another suitable conductive material, or any combination of the foregoing.

As shown in cross-sectional view **900** of FIG. **9**, an etch stop layer **404** is formed along an upper surface of a topmost ILD layer **402** in the lower region **104/r** of the interconnect structure **104**. In some embodiments, the etch stop layer **404** may, for example, be formed by CVD, PVD, ALD, or another suitable growth or deposition process. In further embodiments, the etch stop layer **404** may, for example, be or comprise silicon nitride, silicon carbide, silicon oxynitride, silicon oxycarbide, an extreme low k (ELK) dielectric material, another suitable dielectric material, or any combination of the foregoing. An ELK dielectric material may, for example, be a dielectric having a dielectric constant less than about 2.5, about 2.0, or some other suitable value.

In further embodiments, a process for forming the etch stop layer **404** may include: 1) depositing (e.g., by a CVD process) a dielectric material (e.g., an oxide, such as silicon dioxide, a low-k dielectric material, etc.) over the topmost ILD layer **402** in the lower region **104/r** of the interconnect structure **104**; and 2) performing a planarization process (e.g., a CMP process) into the dielectric material to thin the etch stop layer **404** and to flatten or substantially flatten a top surface **404/ts** of the etch stop layer **404**. In some embodiments, the flattened or substantially flattened top surface is within a tolerance of a CMP process. In some embodiments, the dielectric material is deposited so an initial thickness t_i of the etch stop layer **404** is within a range of about 3,000

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to 5,000 Angstroms before the planarization process. It will be appreciated that other values for the initial thickness t_i are also within the scope of the disclosure. In some embodiments, after the planarization process, the etch stop layer 404 has a thickness t_1 that is about 1,000 Angstroms or within a range of about 100 to 1,000 Angstroms. It will be appreciated that other values for the thickness t_1 are also within the scope of the disclosure. In some embodiments, at any point a height of the top surface 404 t_s of the etch stop layer 404 varies within a range of about -5% to +5% of the thickness t_1 from a level horizontal line 407 located along the top surface 404 t_s of the etch stop layer 404. For example, if the thickness t_1 is about 1,000 Angstroms, then the height of the top surface 404 t_s of the etch stop layer 404 varies within a range of about -50 Angstroms to +50 Angstroms from the level horizontal line 407. In other embodiments, at any point the height of the top surface 404 t_s of the etch stop layer 404 varies within a range of about -5 Angstroms to +5 Angstroms from the level horizontal line 407. It will be appreciated that other values for the height of the top surface 404 t_s of the etch stop layer 404 varying from the level horizontal line 407 are also within the scope of the disclosure.

As shown in cross-sectional view 1000 of FIG. 10, a word line layer 1002 is deposited over the etch stop layer 404, and a threshold selector layer 1004 is deposited along the word line layer 1002. In some embodiments, the word line layer 1002 and/or the threshold selector layer 1004 are deposited by, for example, CVD, PVD, ALD, or another suitable deposition or growth process. In further embodiments, the word line layer 1002 may, for example, be or comprise copper, aluminum, tungsten, titanium nitride, tantalum nitride, another suitable conductive material, or any combination of the foregoing.

In some embodiments, the word line layer 1002 is deposited along the top surface 404 t_s of the etch stop layer 404, thereby ensuring a top surface 1002 t_s of the word line layer 1002 is substantially flat (e.g., a flat top surface within a tolerance of a CMP process). In some embodiments, the word line layer 1002 may comprise a single material, such as tungsten. The single material may have a relatively high hardness, such that the substantially flat top surface 404 t_s of the etch stop layer 404 prevents stress in the word line layer 1002. This, in part, mitigates delamination of the word line layer 1002 and layers and/or structures formed over the word line layer 1002 (e.g., the threshold selector layer 1004).

As shown in cross-sectional view 1100 of FIG. 11, a patterning process is performed on the word line layer (1002 of FIG. 10) and the threshold selector layer (1004 of FIG. 10), thereby defining a word line(s) 120 and threshold selector line(s) 1102. In some embodiments, the patterning process includes: 1) forming a masking layer (not shown) over the threshold selector layer (1004 of FIG. 10); 2) performing a dry etch process on the threshold selector layer (1004 of FIG. 10) and the word line layer (1002 of FIG. 10) according to the masking layer, thereby defining the word line(s) 120 and the threshold selector line(s); 3) and performing a removal process to remove the masking layer. In some embodiments, the word line layer (1002 of FIG. 10) is patterned such that a plurality of word lines 120 are formed that each extend laterally in a first direction (e.g., along the x-axis), as illustrated and described in FIGS. 1-3. In such embodiments, the word lines 120 are arranged in parallel with one another.

In some embodiments, the dry etch process includes performing a reactive-ion etch with a plasma. Due to a

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power of the reactive-ion etch and/or the plasma, charge carriers 1104 (e.g., electrons) may be injected into the word lines 120. Due to a relatively long length of the word lines 120, a large amount of charge carriers 1104 may build up in each word line 120. Because the conductive wires 118 and conductive vias 116 within the lower region 104 r of the interconnect structure 104 are offset from the bottom surface 120 b_s of the word lines 120, the word lines 120 are electrically isolated from the semiconductor devices 106. This prevents the large amount of charge carriers 1104 within the word lines 120 from traveling to the semiconductor devices 106, thereby preventing breakdown of the semiconductor devices 106. In addition, because the conductive vias and wires 116, 118 within the lower region 104 r are offset from the bottom surface 120 b_s of the word lines 120, galvanic corrosion may not occur between the word lines 120 and the conductive vias and wires 116, 118. This further mitigates delamination of the word lines 120 and layers and/or structures subsequently formed over the word lines 120.

As shown in cross-sectional view 1200 of FIG. 12, a dielectric structure 1202 is formed around the word line 120 and the threshold selector line 1102. In some embodiments, a process for forming the dielectric structure 1202 includes: 1) depositing (e.g., by CVD, PVD, ALD, etc.) a dielectric material (e.g., silicon dioxide, a low-k dielectric material, or the like) over the word line 120 and the threshold selector line 1102; and 2) performing a planarization process (e.g., a CMP process) into the dielectric material until an upper surface of the threshold selector line 1102 is reached, thereby defining the dielectric structure 1202.

As shown in cross-sectional view 1300 of FIG. 13, a memory cell film 1302 is formed over the threshold selector line 1102, and a bit line layer 1304 is formed over the memory cell film 1302. In some embodiments, the memory cell film 1302 may be formed by, for example, one or more deposition processes such as PVD, ALD, sputtering, CVD, electroless plating, electroplating, or another suitable deposition or growth process. In further embodiments, the bit line layer 1304 may be formed, for example, by CVD, PVD, ALD, sputtering, or another suitable deposition or growth process. In some embodiments, the bit line layer 1304 may, for example, be or comprise aluminum, copper, titanium nitride, tantalum nitride, tungsten, another suitable conductive material, or any combination of the foregoing.

As shown in cross-sectional view 1400 of FIG. 14, a patterning process is performed on the structure of FIG. 13, thereby defining a plurality of one selector-one memory cell (1S1MC) stacks 126 and a plurality of bit lines 128. In some embodiments, FIG. 4B illustrates alternative embodiments of an orthogonal view of the cross-sectional view 1400 of FIG. 14. For example, FIG. 3 may illustrate a top view of FIG. 14, such that the cross-sectional view 1400 of FIG. 14 is taken along the line A-A' of FIG. 3, and the cross-sectional view of FIG. 4B may be taken along the line B-B' of FIG. 3. Each 1S1MC stack 126 includes a memory cell 124 overlying a threshold selector 122. In some embodiments, the 1S1MC stacks 126 include memory cell layers (e.g., 409 of FIG. 4B), each with multiple memory cells 124, and threshold selectors 122 underlying the memory cell layer (e.g., 409 of FIG. 4B). The plurality of 1S1MC stacks 126 are disposed between the word line 120 and the plurality of bit lines 128. The plurality of bit lines 128 each extend laterally in a second direction (e.g., along the y-axis) transverse the first direction. In some embodiments, the patterning process includes: 1) forming a masking layer (not shown) over the bit line layer (1304 of FIG. 13); 2) exposing

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unmasked regions of the bit line layer (**1304** of FIG. **13**), the memory cell film (**1302** of FIG. **13**), the threshold selector line (**1102** of FIG. **13**), and the dielectric structure **1202** to one or more etchants, thereby defining the plurality of 1S1MC stacks **126** and the plurality of bit lines **128**; and 3) performing a removal process to remove the masking layer.

In further embodiments, the patterning process performed on the structure of FIG. **13** may include performing a reactive-ion etch with a plasma. Due to a power of the reactive-ion etch and/or the plasma, additional charge carriers **1104** may be injected into the word lines **120**.

As shown in cross-sectional view **1500** of FIG. **15**, an upper ILD layer **402u** is formed over the plurality of 1S1MC stacks **126**. In some embodiments, a process for forming the upper ILD layer **402u** includes: 1) depositing (e.g., by PVD, CVD, etc.) a dielectric material (e.g., silicon dioxide, a low-k dielectric material, etc.) over the 1S1MC stacks **126**; and 2) performing a planarization process (e.g., a CMP process) into the dielectric material, thereby defining the upper ILD layer **402u**. In some embodiments, the upper ILD layer **402u** is formed between adjacent 1S1MC stacks **126**, thereby facilitating electrical isolation between adjacent 1S1MC stacks **126**.

In further embodiments, before forming the upper ILD layer **402u**, a charge carrier removal process is performed on the word lines **120** to remove the charge carriers (**1104** of FIG. **14**) from the word lines **120**. In some embodiments, the charge carrier removal process may include exposing the word lines **120** to ultra-violet (UV) light to eject the charge carriers (**1104** of FIG. **14**) from the word lines **120**, electrically coupling the word lines **120** to ground such that the charge carriers (**1104** of FIG. **14**) are removed from the word lines **120**, another suitable charge carrier removal process, or any combination of the foregoing. In some embodiments, exposing the word lines **120** to UV light may remove charge carriers (**1104** of FIG. **14**) from the word lines **120** through the photoelectric effect.

As shown in cross-sectional view **1600** of FIG. **16**, a patterning process is performed on the upper ILD layer **402u** and the etch stop layer **404** to define a plurality of conductive feature openings **1602**. In some embodiments, the patterning process includes: 1) forming a masking layer (not shown) over an upper surface of the upper ILD layer **402u**; 2) exposing unmasked regions of the upper ILD layer **402u** and the etch stop layer **404** to one or more etchants, thereby defining the plurality of conductive feature openings **1602**; and 3) performing a removal process to remove the masking layer. In some embodiments, the plurality of conductive feature openings **1602** each include a conductive wire opening overlying a conductive via opening. Further, a conductive feature opening **1602** exposes at least a portion of a top surface **120ts** of the word line **120**.

As shown in cross-sectional view **1700** of FIG. **17**, a plurality of upper conductive vias **116u** and a plurality of upper conductive wires **118u** are formed within the conductive feature openings (**1602** of FIG. **16**). In some embodiments, a process for forming the upper conductive vias and wires **116u**, **118u** includes: 1) depositing (e.g., by CVD, PVD, sputtering, electroless plating, electroplating, etc.) a conductive structure within the conductive feature openings (**1602** of FIG. **16**); and 2) performing a planarization process (e.g., a CMP process) into the conductive structure until an upper surface of the upper ILD layer **402u** is reached, thereby defining the upper conductive vias and wires **116u**, **118u**. In further embodiments, an upper conductive via **116u** directly overlies the top surface **120ts** of the word line **120** and extends continuously from an upper conductive wire

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118u to abut the top surface **120ts** of the word line **120**. In some embodiments, the upper conductive via **116u** directly contacts the top surface **120ts** of the word line **120**. In yet further embodiments, the upper conductive wires **118u** are electrically coupled to the conductive vias and/or wires **116**, **118** disposed within the lower region **104lr** of the interconnect structure **104**. Further, in some embodiments, an upper conductive via **116u** abuts an upper surface of each bit line **128** (not shown) (e.g., see FIGS. **2-3**). By virtue of the upper conductive vias **116u** being formed after the word line **120**, a conductive path is formed between the word line **120** and a corresponding semiconductor device **106** after forming the bit lines **128**. In some embodiments, charge carriers (e.g., electrons) accumulated in the word line **120** may be dispersed before forming the conductive path (e.g., by performing the charge carrier removal process, described in FIG. **15**, before forming the conductive path). This reduces breakdown of the semiconductor devices **106**.

As shown in cross-sectional view **1800** of FIG. **18**, a first passivation layer **406** is formed over the upper ILD layer **402u**, and a plurality of bond pads **405** are formed within the first passivation layer **406**. The bond pads **405** are formed over a corresponding upper conductive wire **118u**. In some embodiments, the first passivation layer **406** may, for example, be deposited by CVD, PVD, ALD, or another suitable deposition process. In further embodiments, the bond pads **405** may be formed, for example, by CVD, PVD, sputtering, electroless plating, electroplating, or another suitable deposition or growth process.

As shown in cross-sectional view **1900** of FIG. **19**, a second passivation layer **408** is formed over the first passivation layer **406**. In some embodiments, the second passivation layer **408** may be formed, for example, by CVD, PVD, ALD, or another suitable deposition process.

FIG. **20** illustrates a method **2000** of some embodiments for forming a memory device having a plurality of memory cells disposed between overlying bit lines and underlying word lines in an interconnect structure, where an upper conductive via extends from above the bit lines to a top surface of a corresponding word line according to the present disclosure. Although the method **2000** is illustrated and/or described as a series of acts or events, it will be appreciated that the method is not limited to the illustrated ordering or acts. Thus, in some embodiments, the acts may be carried out in different orders than illustrated, and/or may be carried out concurrently. Further, in some embodiments, the illustrated acts or events may be subdivided into multiple acts or events, which may be carried out at separate times or concurrently with other acts or sub-acts. In some embodiments, some illustrated acts or events may be omitted and other un-illustrated acts or events may be included.

At act **2002**, semiconductor devices are formed within and/or over a substrate. FIG. **8** illustrates a cross-sectional view **800** corresponding to some embodiments of act **2002**.

At act **2004**, a lower region of an interconnect structure is formed over the semiconductor devices and the substrate. The interconnect structure comprises a plurality of conductive vias and a plurality of conductive wires. FIG. **8** illustrates a cross-sectional view **800** corresponding to some embodiments of act **2004**.

At act **2006**, an etch stop layer is formed along an upper surface of the lower region of the interconnect structure. FIG. **9** illustrates a cross-sectional view **900** corresponding to some embodiments of act **2006**.

At act **2008**, a word line layer is formed over the etch stop layer, and a threshold selector layer is formed over the word line layer. The conductive vias and wires are offset and/or

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completely spaced from a bottom surface of the word line layer. FIG. 10 illustrates a cross-sectional view 1000 corresponding to some embodiments of act 2008.

At act 2010, the word line layer and the threshold selector layer are patterned to define a word line underlying a threshold selector line. FIG. 11 illustrates a cross-sectional view 1100 corresponding to some embodiments of act 2010.

At act 2012, a memory cell film is formed over the threshold selector line, and a bit line layer is formed over the memory cell film. FIG. 13 illustrates a cross-sectional view 1300 corresponding to some embodiments of act 2012.

At act 2014, the bit line layer, the memory cell film, and the threshold selector line are patterned, thereby defining a plurality of bit lines and a plurality of one selector-one memory cell (1S1MC) stacks. FIG. 14 illustrates a cross-sectional view 1400 corresponding to some embodiments of act 2014.

At act 2016, an upper inter-level dielectric (ILD) layer is formed over the plurality of 1S1MC stacks. FIG. 15 illustrates a cross-sectional view 1500 corresponding to some embodiments of act 2016.

At act 2018, the upper ILD layer is patterned to define a plurality of conductive feature openings within the upper ILD layer. A conductive feature opening overlies and exposes a top surface of the word line. FIG. 16 illustrates a cross-sectional view 1600 corresponding to some embodiments of act 2018.

At act 2020, a plurality of upper conductive vias and upper conductive wires are formed within the conductive feature openings. An upper conductive via overlies and abuts the top surface of the word line and partially defines a conductive path to at least one of the semiconductor devices. FIG. 17 illustrates a cross-sectional view 1700 corresponding to some embodiments of act 2020.

Accordingly, in some embodiments, the present disclosure is related towards a memory device having a plurality of memory cells disposed between overlying bit lines and underlying word lines in an interconnect structure, where an upper conductive via extends from above the memory cells to a top surface of a corresponding word line.

In some embodiments, the present application provides an integrated chip including: a substrate; an array overlying the substrate and comprising multiple memory stacks in a plurality of rows and a plurality of columns, wherein each of the memory stacks comprises a data storage structure having a variable resistance; a plurality of word lines disposed beneath the array and extending along corresponding rows of the array, wherein the word lines are electrically coupled with memory stacks of the array in the corresponding rows; and a plurality of upper conductive vias extending from above the array of memory stacks to contact top surfaces of corresponding word lines. In an embodiment, the integrated chip further includes an interconnect dielectric structure overlying the substrate, wherein the memory stacks, the word lines, and the upper conductive vias are disposed within the interconnect dielectric structure, and wherein the interconnect dielectric structure directly contacts a bottom surface of each word line across an entirety of the bottom surface. In an embodiment, the bottom surface of each word line is defined between a first outer sidewall of a corresponding word line and a second outer sidewall of the corresponding word line, wherein the first outer sidewall is opposite the second outer sidewall, and wherein the interconnect dielectric structure continuously laterally extends along an unbroken path from the first outer sidewall to the second outer sidewall. In an embodiment, the integrated chip further includes multiple bit lines disposed above the array

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and extending along corresponding columns of the array, wherein the bit lines are electrically coupled with memory stacks of the array in the corresponding columns. In an embodiment, a top surface of the plurality of upper conductive vias is disposed above the bit lines. In an embodiment, the memory stacks include corresponding memory cells overlying corresponding threshold selectors. In an embodiment, a bottom surface of one of the threshold selectors is disposed along a top surface of an underlying word line. In an embodiment, a width of the word lines is greater than a width of the memory stacks. In an embodiment, the word lines comprise a first material and the upper conductive vias comprise a second material different than the first material.

In some embodiments, the present application provides an integrated chip including: a substrate; an interconnect structure overlying the substrate, wherein the interconnect structure includes a plurality of conductive wires and a plurality of conductive vias disposed within a dielectric structure, an array of memory stacks disposed within the dielectric structure in a plurality of rows and a plurality of columns, wherein the memory stacks respectively include memory cells overlying selectors; a plurality of bit lines disposed within the dielectric structure and extending along corresponding columns of the array in a first direction, wherein the bit lines overlie the memory stacks and are electrically coupled with memory cells of the array in the corresponding columns; a plurality of word lines disposed within the dielectric structure and extending along corresponding rows of the array in a second direction, wherein the second direction is orthogonal to the first direction, and wherein the word lines underlie the memory stacks and are electrically coupled with memory cells of the array in the corresponding rows; and wherein a bottom surface of each word line is completely separated from the conductive vias and the conductive wires. In an embodiment, the plurality of conductive vias comprise a plurality of upper conductive vias that extend from above the bit lines to directly contact top surfaces of corresponding word lines. In an embodiment, the dielectric structure includes an etch stop layer disposed between the word lines and conductive vias and wires disposed beneath the word lines, and wherein a top surface of the etch stop layer directly contacts an entirety of the bottom surface of each word line. In an embodiment, the top surface of the etch stop layer is substantially flat. In an embodiment, the interconnect structure is devoid of a conductive via directly contacting a bottom surface of the word lines. In an embodiment, the integrated chip further includes: a second array comprising multiple second memory stacks in a plurality of rows and a plurality of columns, wherein the second memory stacks respectively comprise second memory cells overlying second selectors, wherein the bit lines extend along corresponding columns of the second array and electrically couple with second memory cells of the second array in the corresponding columns; and upper word lines extending along corresponding rows of the second array and electrically coupled with second memory cells of the second array in the corresponding rows, wherein the upper word lines extend in the second direction, and wherein the bit lines are disposed between the upper word lines and the word lines. In an embodiment, the conductive vias and wires comprise aluminum and/or copper, and wherein the bit lines and the word lines comprise tungsten.

In some embodiments, the present application provides a method for forming a memory device, the method including: forming a lower interconnect structure over a substrate, wherein the lower interconnect structure includes conduc-

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tive vias and conductive wires disposed within a dielectric structure; forming an etch stop layer along an upper surface of the lower interconnect structure; forming a plurality of word lines over the etch stop layer such that the conductive vias and wires are offset from a bottom surface of the word lines, wherein the word lines extend along a first direction and are parallel to one another; forming a plurality of memory stacks along top surfaces respectively of the word lines; forming a plurality of bit lines over the memory stacks such that the bit lines extend along a second direction orthogonal to the first direction, wherein the bit lines are parallel to one another; forming an upper ILD layer over the bit lines; and forming a plurality of upper conductive vias within the upper ILD layer such that the upper conductive vias contact top surfaces of corresponding word lines, wherein top surfaces respectively of the upper conductive vias are disposed above the bit lines. In an embodiment, the method further include: forming a plurality of semiconductor devices on the substrate, wherein the conductive vias and wires within the lower interconnect structure are electrically coupled to the semiconductor devices; wherein the word lines are electrically isolated from the semiconductor devices before forming the memory stacks and the bit lines, and wherein the word lines are electrically coupled to the semiconductor devices after forming the upper conductive vias. In an embodiment, forming the etch stop layer includes: depositing a dielectric material along a top surface of the lower interconnect structure; and performing a planarization process into the dielectric material, thereby defining the etch stop layer with a substantially flat top surface. In an embodiment, forming the plurality of word lines includes: depositing a word line layer along a top surface of the etch stop layer; and performing a dry etch process on the word line layer to define the plurality of word lines.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. An integrated chip comprising:

- a first conductive interconnect structure overlying a substrate;
- a first memory stack disposed on the first conductive interconnect structure;
- a second conductive interconnect structure overlying the first memory stack, wherein the second conductive interconnect structure is spaced laterally between opposing sidewalls of the first conductive interconnect structure; and
- a third conductive interconnect structure disposed on the first conductive interconnect structure, wherein a top surface of the third conductive interconnect structure is vertically above the second conductive interconnect structure.

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2. The integrated chip of claim 1, further comprising:

- a second memory stack disposed on the first conductive interconnect structure and laterally offset from the first memory stack and the second conductive interconnect structure.

3. The integrated chip of claim 1, wherein a bottom surface of the third conductive interconnect structure is vertically aligned with a bottom surface of the first memory stack.

4. The integrated chip of claim 1, further comprising:

- a semiconductor device disposed on the substrate;
- a fourth conductive interconnect structure over the third conductive interconnect structure; and
- a fifth conductive interconnect structure extending from the fourth conductive interconnect structure in a direction towards the semiconductor device, wherein the first conductive interconnect structure is directly electrically coupled to the semiconductor device by way of the third, fourth, and fifth conductive interconnect structures.

5. The integrated chip of claim 4, wherein a height of the fifth conductive interconnect structure is greater than a height of the third conductive interconnect structure.

6. The integrated chip of claim 1, further comprising:

- a dielectric structure disposed between the substrate and the first conductive interconnect structure, wherein the first conductive interconnect structure comprises a single planar bottom surface, wherein the dielectric structure contacts a total area of the single planar bottom surface.

7. The integrated chip of claim 1, further comprising:

- a fourth conductive interconnect structure disposed over and on the second conductive interconnect structure, wherein the top surface of the third conductive interconnect structure is aligned with a top surface of the fourth conductive interconnect structure.

8. The integrated chip of claim 7, wherein a height of the third conductive interconnect structure is equal to a distance between a top surface of the first conductive interconnect structure and a bottom surface of the fourth conductive interconnect structure.

9. A semiconductor structure comprising:

- a first metal wire disposed over a substrate;
- a first memory structure and a second memory structure disposed on the first metal wire; and
- a first metal via overlying and contacting the first metal wire, wherein the second memory structure is spaced laterally between the first memory structure and the first metal via, and wherein a bottom surface of the first metal via is aligned with a bottom surface of the first memory structure.

10. The semiconductor structure of claim 9, wherein a height of the first metal via is greater than a height of the first memory structure.

11. The semiconductor structure of claim 9, further comprising:

- a lower interconnect structure disposed between the first metal wire and the substrate, wherein the lower interconnect structure comprises a plurality of lower wires and a plurality of lower vias, wherein a bottom surface of the first metal wire is fully spaced from the lower wires and vias by a dielectric structure.

12. The semiconductor structure of claim 11, wherein the first metal wire is directly electrically coupled to one or more of the lower wires and/or lower vias by the first metal via.

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13. The semiconductor structure of claim **9**, wherein a width of the first metal via is less than a width of the first memory structure.

14. The semiconductor structure of claim **9**, further comprising:

a second metal wire overlying the first memory structure, wherein the second metal wire is spaced vertically between a top surface and the bottom surface of the first metal via.

15. The semiconductor structure of claim **14**, further comprising:

a second metal via over the second metal wire and having a top surface coplanar with the top surface of the first metal via.

16. A method for forming an integrated chip comprising: forming a first conductive interconnect structure over a substrate;

forming a memory stack on the first conductive interconnect structure;

forming a second conductive interconnect structure over the memory stack; and

forming a third conductive interconnect structure on the first conductive interconnect structure, wherein the third conductive interconnect structure extends from a point above the second conductive interconnect structure.

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17. The method of claim **16**, wherein a bottom surface of the third conductive interconnect structure is aligned with a bottom surface of the memory stack.

18. The method of claim **16**, further comprising:

forming a lower interconnect structure over the substrate, wherein the lower interconnect structure comprises a plurality of lower conductive interconnect structures disposed within a dielectric structure; and

forming a dielectric layer over the lower interconnect structure, wherein the first conductive interconnect structure is formed on a top surface of the dielectric layer, wherein the first conductive interconnect structure is vertically separated from surfaces of the lower conductive interconnect structures by at least a thickness of the dielectric layer.

19. The method of claim **18**, wherein before forming the third conductive interconnect structure the first conductive interconnect structure is electrically isolated from the plurality of lower conductive interconnect structures.

20. The method of claim **16**, wherein the first conductive interconnect structure comprises a first material and the third conductive interconnect structure comprises a second material different from the first material.

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